FACE DETECTION

Project work submitted to Thiruvalluvar University, Serkadu, Vellore.

In partial fulfillment for the award of degree

A CRM APPLICATION FOR THE CHAI SHOP & CO

BY

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SHANMUGA INDUSTRIES ARTS & SCIENCE COLLEGE

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A CRM APPLICATION FOR THECHAISHOP&CO

User Story:

As a sales representative at TheChaiShop&Co, I want to efficiently manage customer relationships and streamline sales processes using our CRM application.
I want to easily access and update employer details, as well as distinguish between online and offline customer records using record types, allowing for personalized interactions.
Additionally, I need the application to automate sales calculations, such as order quantity and total bill amount, to ensure accurate and efficient transactions.
Moreover, I require the ability to generate insightful reports and dashboards to summarize and analyze sales data effectively, aiding in decision-making and strategy development.
Lastly, I would like the application to automate email notifications to customers on their birthdays, informing them of special offers tailored to celebrate their special day, enhancing customer engagement and loyalty.

1. INTRODUCTION

1.1 Project Overview :

The project “A CRM Application for TheChaiShop&Co” focuses on developing a Salesforce-based Customer Relationship Management (CRM) system to manage customer, employer, and enterprise accounts efficiently. The system automates sales processes, maintains customer records, generates reports, and improves customer engagement through automated notifications.

The project user story and objectives are defined in the initial section of the document

1.2 Purpose :

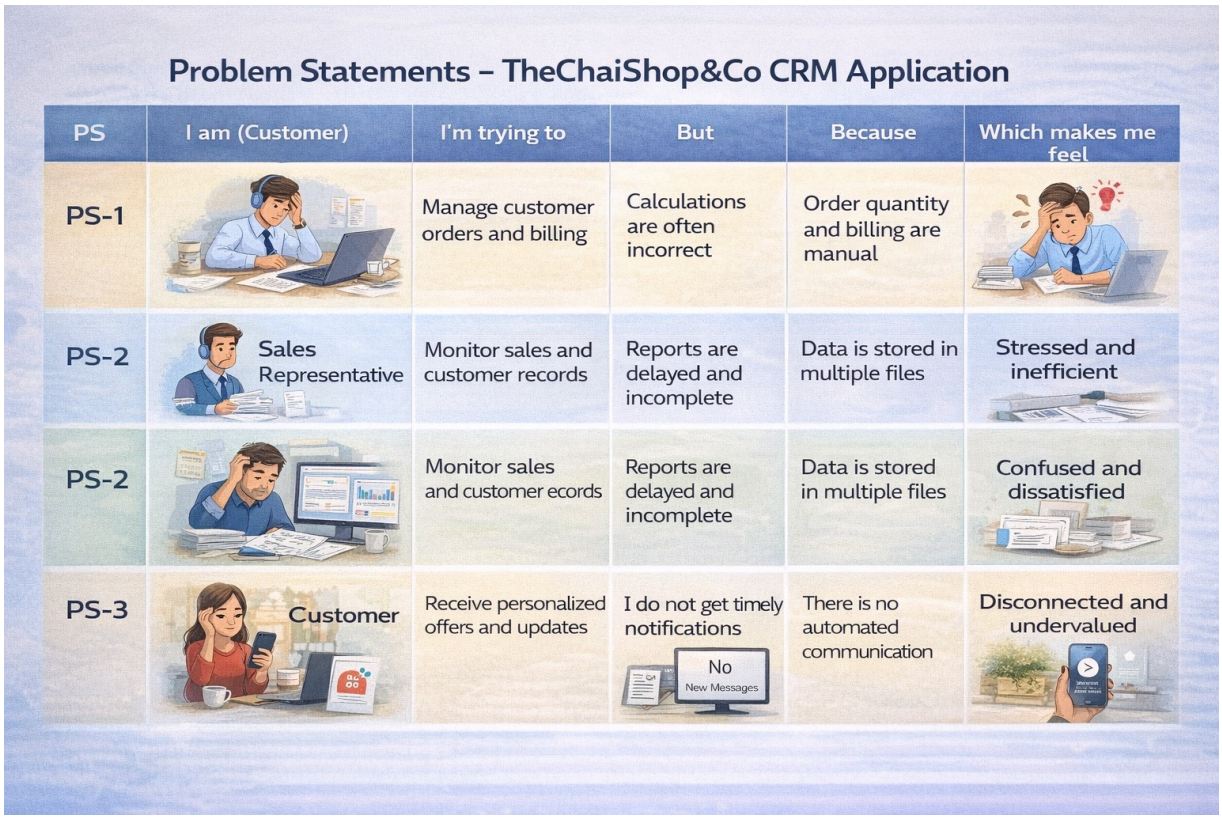
The purpose of this project is to:

- Streamline customer relationship management
- Automate billing and order calculations
- Improve data accuracy
- Enable data-driven decision-making
- Enhance customer engagement through email automation

2. IDEATION PHASE

2.1 Problem Statement (Customer Problem Statement Template)

PS	**I am (Customer)**	**I'm trying to**	**But**	**Because**	**Which makes me feel**
PS-1	Sales Representative	Manage customer orders and billing	Calculations are often incorrect	Order quantity and billing are manual	Stressed and inefficient
PS-2	Store Manager	Monitor sales and customer records	Reports are delayed and incomplete	Data is stored in multiple files	Confused and dissatisfied
PS-3	Customer	Receive personalized offers and updates	I do not get timely notifications	There is no automated communication	Disconnected and undervalued



2.2 Empathy Map Canvas

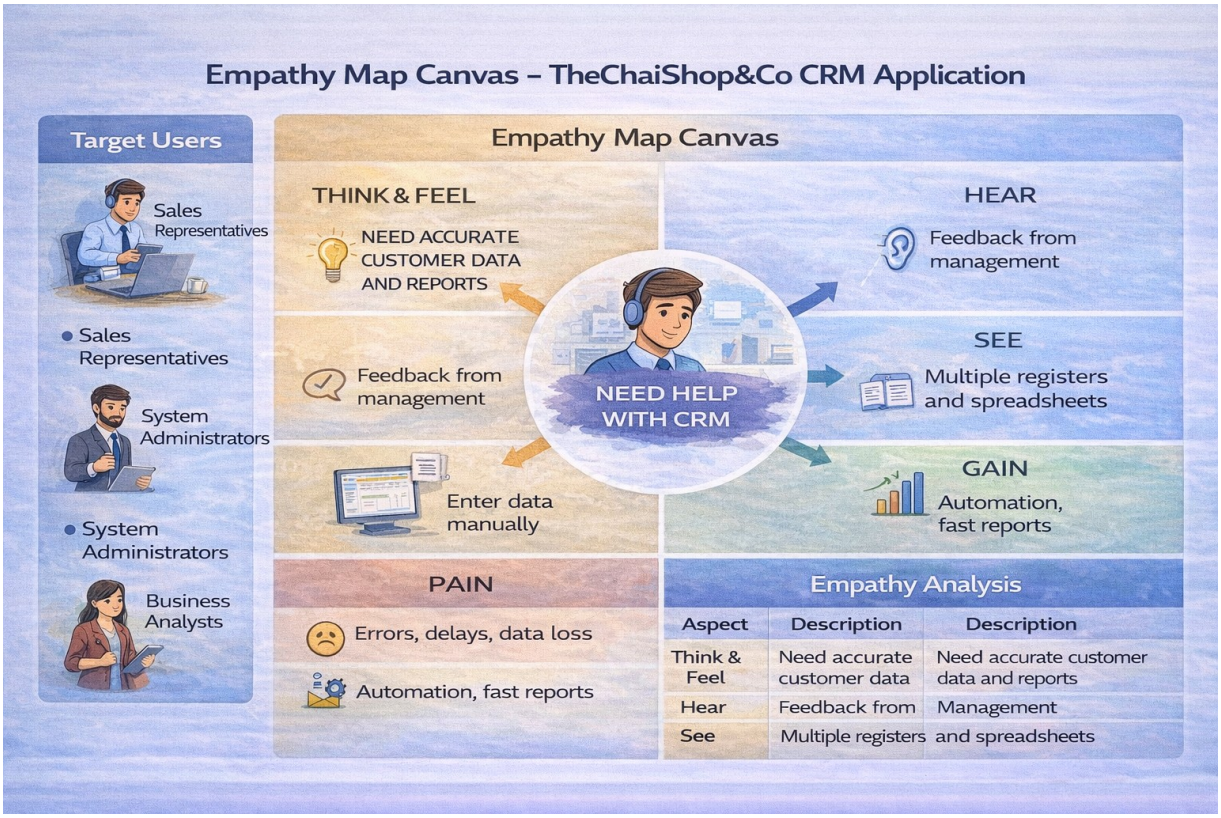
Target Users

- Sales Representatives
- Managers
- System Administrators
- Business Analysts

Empathy Analysis

Aspect	Description
Think & Feel	Need accurate customer data and reports
Hear	Feedback from management

See	Multiple registers and spreadsheets
Say & Do	Enter data manually
Pain	Errors, delays, data loss
Gain	Automation, fast reports



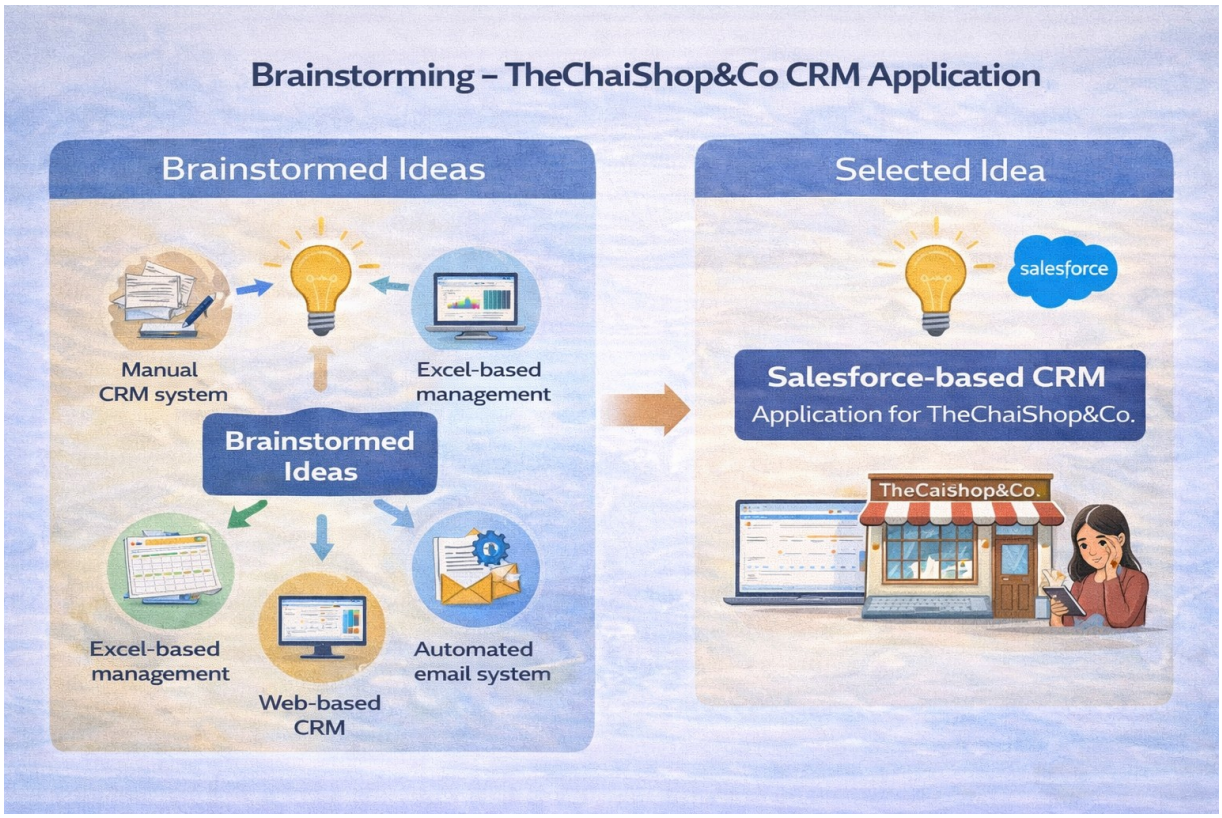
2.3 Brainstorming

Brainstormed Ideas

- Manual CRM system
- Excel-based management
- Web-based CRM
- Salesforce-based CRM
- Automated email system

Selected Idea:

Salesforce-based CRM Application for TheChaiShop&Co..

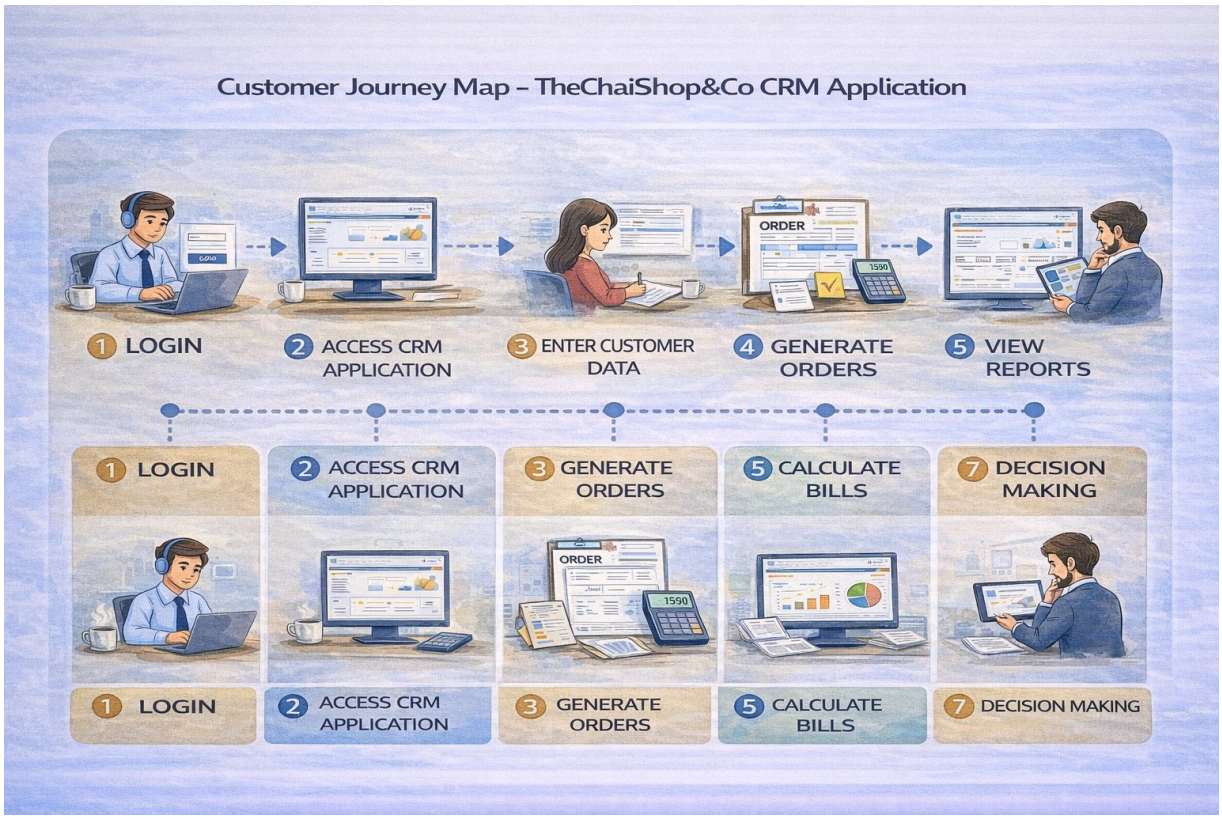


3.REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey Map

Journey Stages:

- Login
- Access CRM Application
- Enter Customer Data
- Generate Orders
- Calculate Bills
- View Reports
- Decision Making



3.2 Solution Requirements

Functional Requirements

FR No	Requirement
FR-1	User Login
FR-2	Object Creation
FR-3	Data Entry
FR-4	Billing Automation
FR-5	Report Generation
FR-6	Email Automation

Non-Functional Requirements

NFR-1: Usability

The system should be easy to use for sales representatives, managers, administrators, and analysts without requiring technical training.

Requirements :

- Simple and intuitive user interface
- Clear navigation menus
- Easy-to-understand forms
- User-friendly dashboards

Example:

A new sales employee should be able to log in, enter customer details, and generate orders within a few minutes without help.

Salesforce Lightning UI provides drag-and-drop features and easy navigation.

NFR-2: Security

The system must protect customer and business data from unauthorized access.

Requirements:

- Role-based access control
- Secure login authentication
- Encrypted data storage
- Data backup and recovery

Examples:

- Profiles and permission sets
- Object-level and field-level security
- Restricted billing access
- Secure login and password policies

NFR-3: Reliability

The CRM system should work correctly at all times and store data accurately without loss.

Requirements:

- Automatic data saving
- Regular system backups
- Error handling mechanisms
- Stable cloud environment

Example :

If a user enters order data and the internet disconnects, the system should save the information and restore it later.

Salesforce cloud ensures automatic backups.

NFR-4: Performance

The system should respond quickly and handle large amounts of data efficiently.

Requirements:

- Fast page loading
- Quick report generation
- Smooth dashboard rendering
- Minimal delay

Example: When a manager generates monthly sales reports for hundreds of customers, the report should load within 2–3 seconds.

Optimized queries and caching improve speed.

NFR-5: Availability

The system should be accessible whenever required.

Requirements:

- 24/7 system availability
- Minimal downtime
- Cloud-based hosting
- Disaster recovery

Example: Managers should be able to access the CRM system during peak business hours, weekends, or late nights.

Salesforce cloud provides high uptime.

NFR-6: Scalability

The system should support business growth without affecting performance.

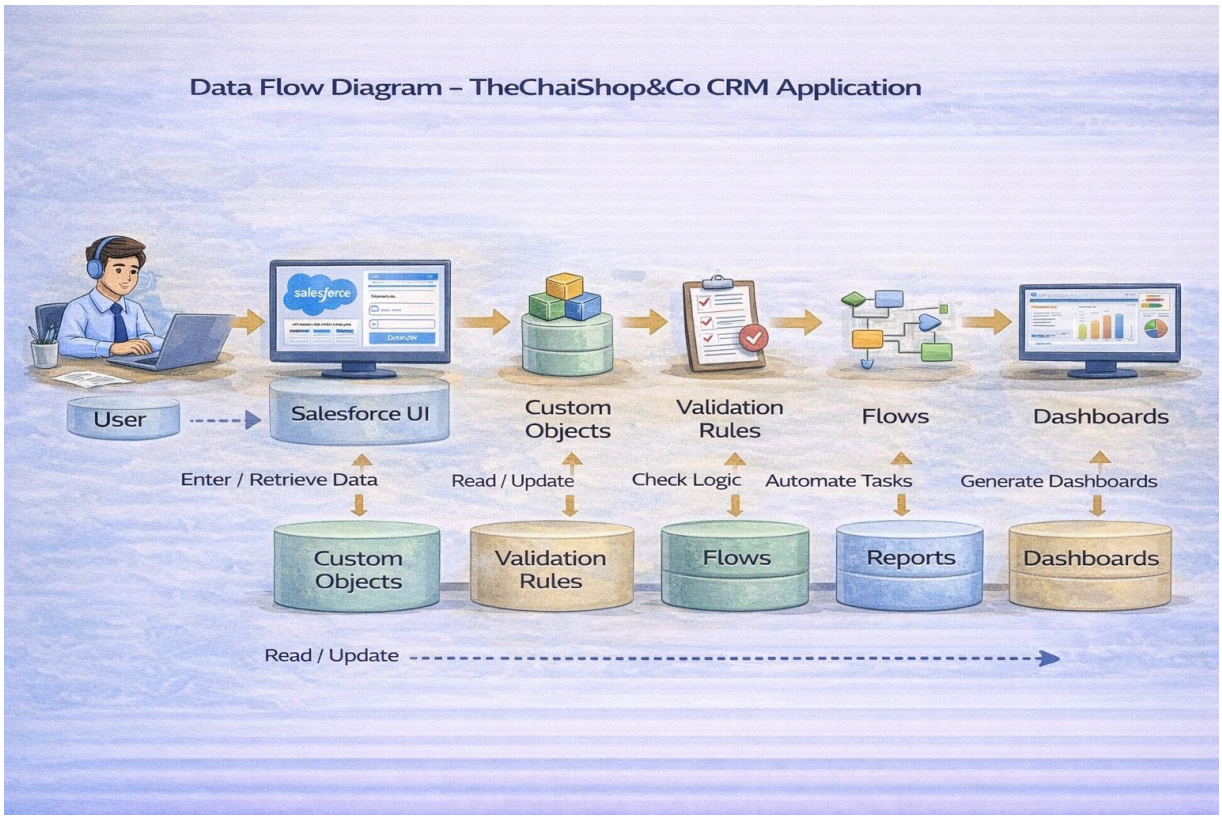
Requirements:

- Ability to handle more users
- Support increasing customer data
- Expandable storage
- Maintain system speed

Example: If TheChaiShop&Co expands to multiple branches, the CRM system should support thousands of customers without slowing down.
Salesforce infrastructure automatically scales.

3.3 Data Flow Diagram

User → Salesforce UI → Custom Objects → Validation Rules → Flows → Reports → Dashboards.



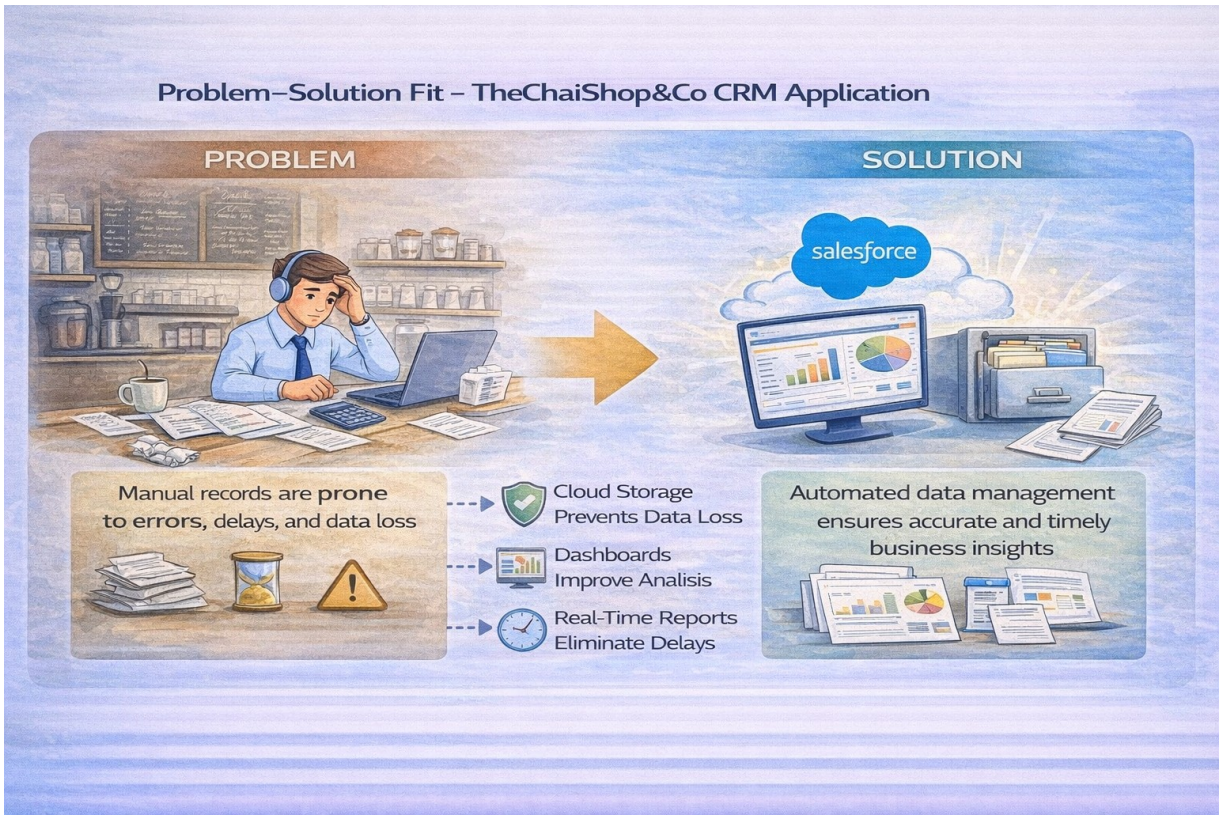
3.4 Technology Stack

Layer	Technology
Frontend	Salesforce Lightning
Database	Salesforce Cloud
Automation	Flows
Backend	Salesforce Platform
Analytics	Reports & Dashboards

4. PROJECT DESIGN PHASE

4.1 Problem–Solution Fit

The CRM system replaces manual records with automated data management. Cloud storage prevents data loss, dashboards improve analysis, and real-time reports eliminate delays. This ensures accurate and timely business insights generation.



4.2 Proposed Solution

Functional Requirements

Parameter	Description
User Login & Authentication	Allows authorized users to securely log in to the CRM system using valid credentials.
Customer Record Management	Enables users to create, view, update, and delete customer details such as name, contact number, and address.
Employer & Enterprise Management	Allows management of employer and enterprise account information for business operations.

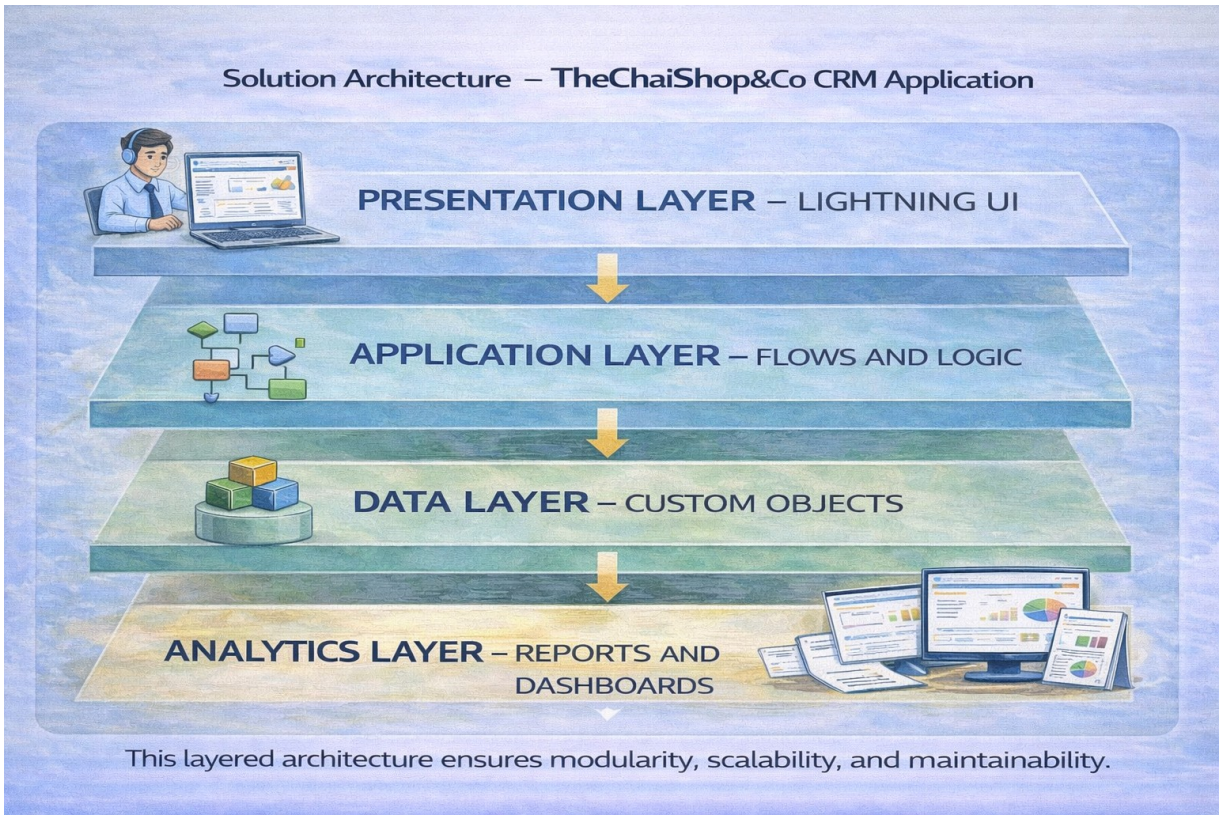
Order Management	Supports creation and tracking of customer orders, including order quantity and product details.
Automated Billing System	Automatically calculates total bill amount using formula fields based on order quantity and pricing.
Validation Rules	Ensures accurate data entry by validating fields such as phone number, PAN, and Aadhaar.
Record Types & Page Layouts	Provides different record types and layouts for various customer and enterprise categories.
Global Picklists	Maintains standardized values for fields such as product type, category, and status.
Email Notification System	Sends automated emails to customers for birthdays, offers, and order confirmations.
Workflow & Flow Automation	Automates tasks such as updating records and triggering alerts using Salesforce Flows.
Report Generation	Generates detailed reports on sales, customers, and enterprise performance.
Dashboard Visualization	Displays real-time analytics through interactive dashboards and charts.
Data Backup & Recovery	Ensures secure storage and recovery of business data in the Salesforce cloud.
User Role Management	Controls system access using profiles and permission sets for different users.

4.3 Solution Architecture

Architecture Layers

1. Presentation Layer – Lightning UI
2. Application Layer – Flows and Logic
3. Data Layer – Custom Objects
4. Analytics Layer – Reports and Dashboards

This layered architecture ensures modularity, scalability, and maintainability.



5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

- The project followed an Agile-based development approach with defined milestones and sprints.
- A Salesforce Developer account was created to set up the development environment.
- Custom objects were designed to store customer, employer, and enterprise data.
- Custom tabs and a Lightning Application were developed for easy system navigation.
- Fields, relationships, and global picklists were configured to ensure data consistency.
- Page layouts and record types were created based on user roles and customer categories.
- Validation rules were implemented to prevent incorrect data entry.
- Email templates and automated flows were developed for customer notifications.

- Reports and dashboards were created for performance monitoring and analysis.
- Regular sprint reviews and testing were conducted to track progress and resolve issues.
- The project was completed within the planned timeline through systematic scheduling.

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Developer Setup	USN-1	As an admin, I want to create a Salesforce developer account so that I can configure and deploy the TheChaiShop&Co CRM application.	1	High	Member1
Sprint-1	Data Modeling	USN-2	As an admin, I want to create Customer, Employer, and Enterprise objects so that business data can be stored and managed systematically.	3	High	Member2
Sprint-2	Data Modeling	USN-3	As a user, I want separate tabs for each object so that I can easily navigate and manage CRM records within the application.	3	High	Member3
Sprint-2	Data Modeling	USN-4	As a manager, I want a dedicated Lightning App so that all CRM operations are accessible from a single dashboard.	3	High	Member4
Sprint-2	Data Modeling	USN-5	As an admin, I want to create customer and order fields so that complete sales and billing details can be captured accurately.	5	Medium	Member1
Sprint-2	Data Modeling	USN-6	As a user, I want customized page layouts for different record types so that only relevant information is displayed during data entry	5	High	Member2
Sprint-2	Data Modeling	USN-7	As a manager, I want separate record types for Retail and Enterprise customers so that different business categories can be managed efficiently.	5	Medium	Member3
Sprint-3	Automation	USN-8	As a system, I want to validate customer details and order inputs so that incorrect data entries are	3	High	Member4

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
			prevented.			
Sprint-3	Automation	USN-9	As a system, I want automated flows to send email notifications so that customers receive timely updates and offers.	5	High	Member1
Sprint-4	Automation	USN-10	As a system, I want to automatically calculate billing amounts so that invoices are generated without manual effort.	5	High	Member2
Sprint-5	Reports	USN-11	As a manager, I want sales and customer reports so that I can monitor business performance and make informed decisions.	4	High	Member3
Sprint-6	Dashboards	USN-12	As a manager, I want visual dashboards so that I can quickly analyze sales, customers, and enterprise performance.	4	Medium	Member4

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	20	6 Days	24 Oct 2022	29 Oct 2022	20	29 Oct 2022
Sprint-2	20	6 Days	31 Oct 2022	05 Nov 2022		
Sprint-3	20	6 Days	07 Nov 2022	12 Nov 2022		
Sprint-4	20	6 Days	14 Nov 2022	19 Nov 2022		

6. Project Development Phase:

Project Flow

Milestone 1 : Creation of developer account
Milestone 2 : Object Creation
Milestone 3 : Tabs
Milestone 4 : The Lightning App
Milestone 5 : Global Picklist
Milestone 6 : Fields and Relationships
Milestone 7 : Creation of Page Layout
Milestone 8 : Creation of Record Types
Milestone 9 : Validation rules
Milestone 10 : Email Templates
Milestone 11 : Flows
Milestone 12 : Reports and Dashboards
Milestone 13 : Conclusion

What you'll learn

1. Real Time Salesforce Project
2. Standard and Custom objects
3. Picklist Value sets
4. Types of Fields & their relationships.
5. Page Layouts
6. Record Types
7. Validation Rules
8. Email Templates
9. Flows
10. Reports and Dashboards.

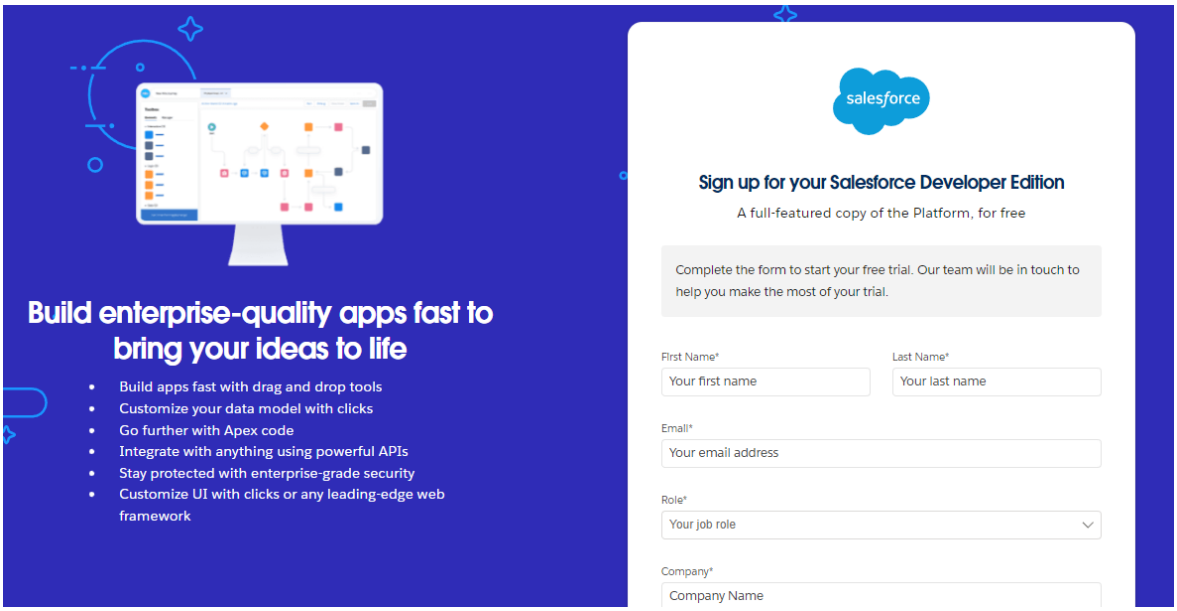
Implementation

Milestone 1 - Salesforce developer account creation :

Activity 1: Creating Developer Account:

Creating a developer org in salesforce.

1. Go to https://developer.salesforce.com/signup
2. On the sign up form, enter the following details :



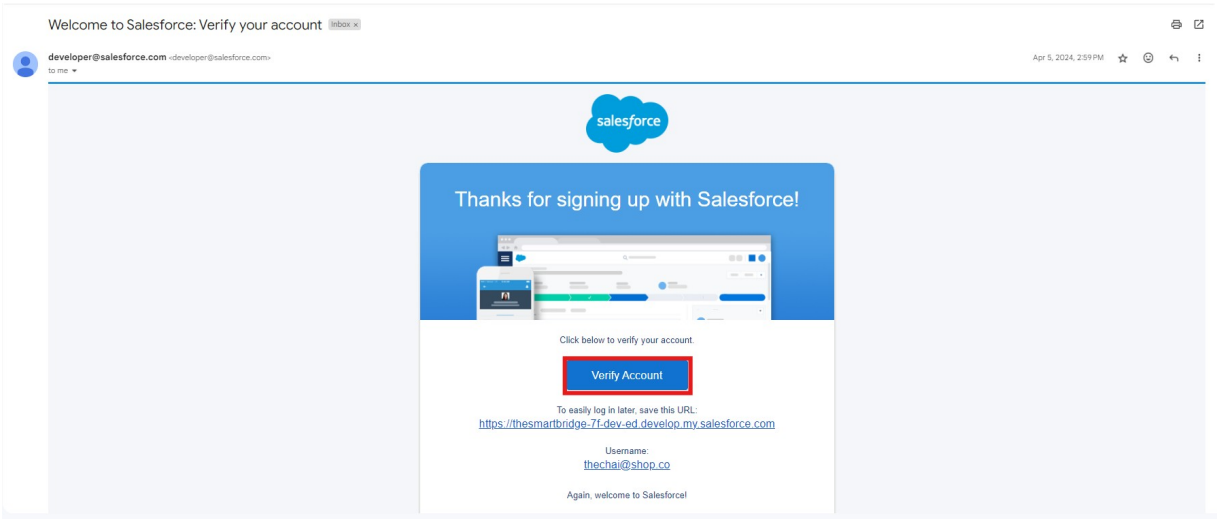
1) First name & Last name
2) Email
3) Role : Developer
4) Company : College or Company Name
5) County : India
6) Postal Code : pin code
7) Username : should be a combination of your name and company

This need not be an actual email id, you can give anything in that format : username@organization.com

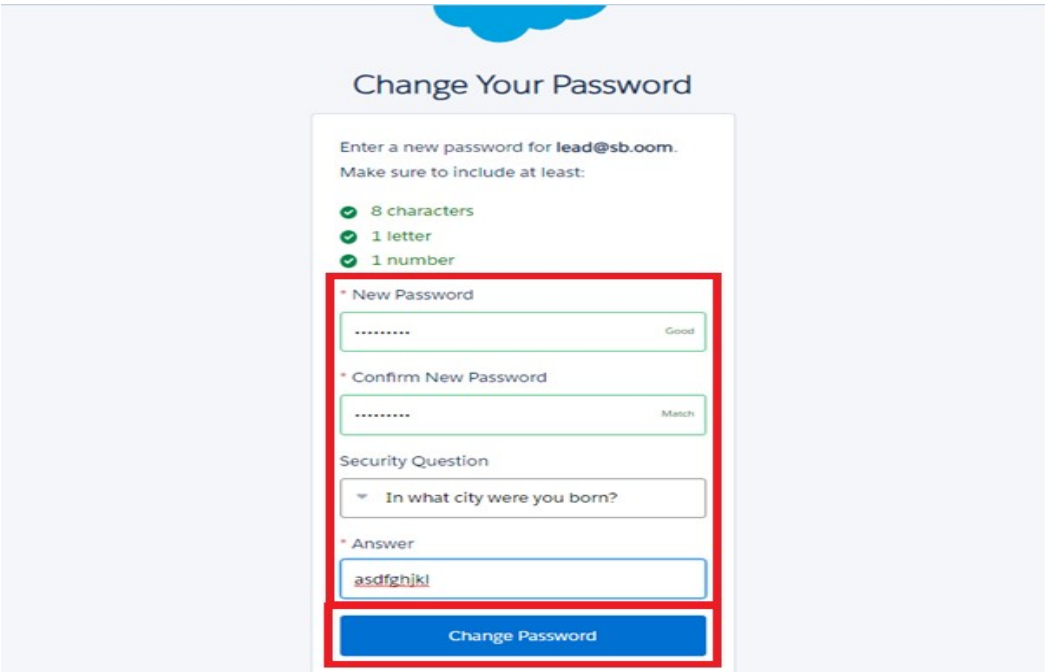
Click on sign me up after filling these.

Activity 2: Account Activation :

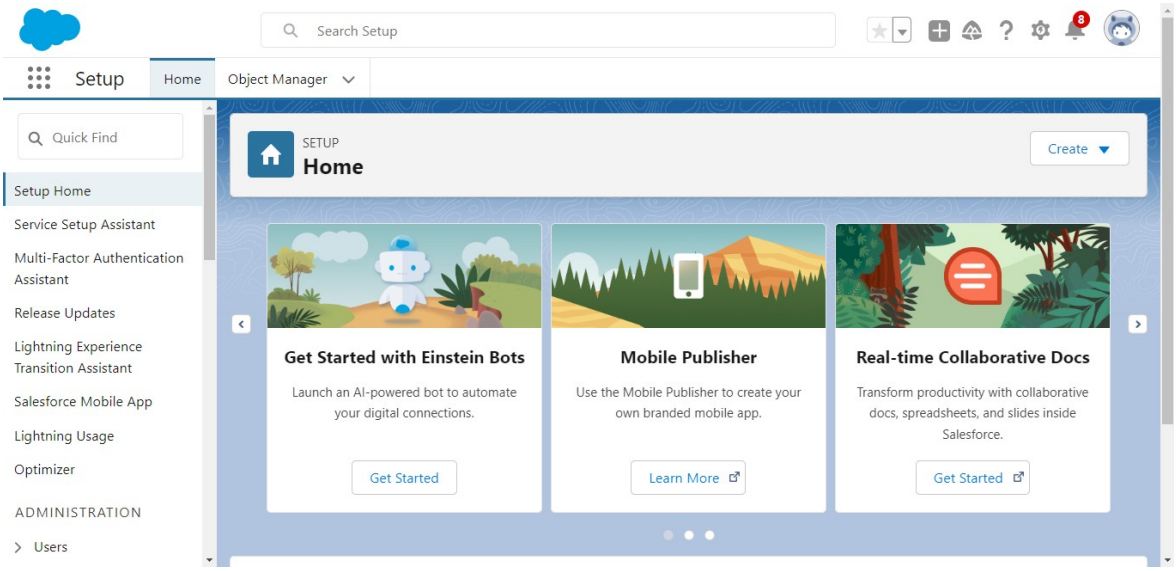
1. Go to the inbox of the email that you used while signing up. Click on the verify account to activate your account. The email may take 5-10mins.



2. Click on Verify Account.
3. Give a password and answer a security question and click on change password.

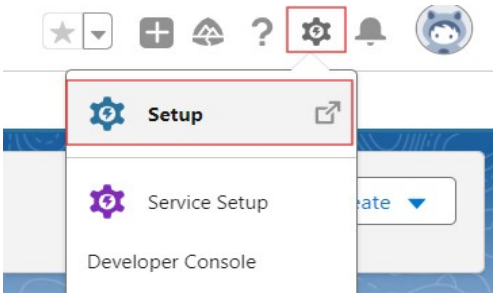


4. Then you will redirect to your salesforce setup page.

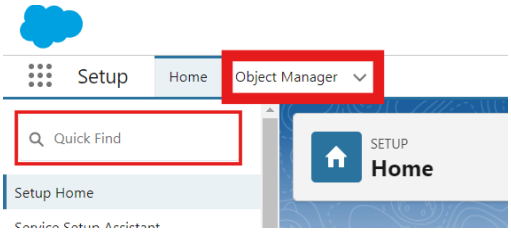


Milestone 2 - Objects

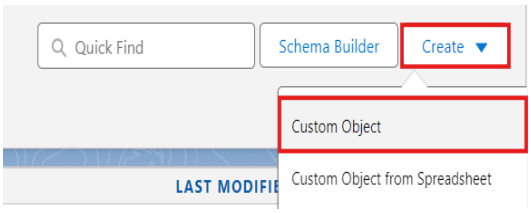
1. In your Salesforce org, click gear icon on the top left and select Setup to open Setup.



2. Click the Object Manager tab. If you don't see it, enter Object Manager in the Quick Find box.



3. On the Object Manager page, click Create | Custom Object.



Activity 1 : Employer Account Object

1. From the setup page → Click on Object Manager → Click on Create → Click on Custom Object.
   - **Label** : Employer Account
   - **Plural Label** : Employer Accounts
   - Enter Record Name Label and Format
     Record Name : Employer Account Name
     Data Type **:** Text
   - **In Optional features :** check the boxes for Allow Reports | Allow Activities | Track Field History.
   - **Search Status :** check the box for Allow Search.
   - Save

Activity 2 : Customer Account Object

1. From the setup page → Click on Object Manager → Click on Create → Click on Custom Object.
    - **Label** : Customer Account
    - **Plural Label** : Customer Accounts
    - Enter Record Name Label and Format
        Record Name : Customer Account Name
        Data Type **:** Text
    - **In Optional features :** check the boxes for Allow Reports | Allow Activities | Track Field History.
    - **Search Status :** check the box for Allow Search.
    - Save

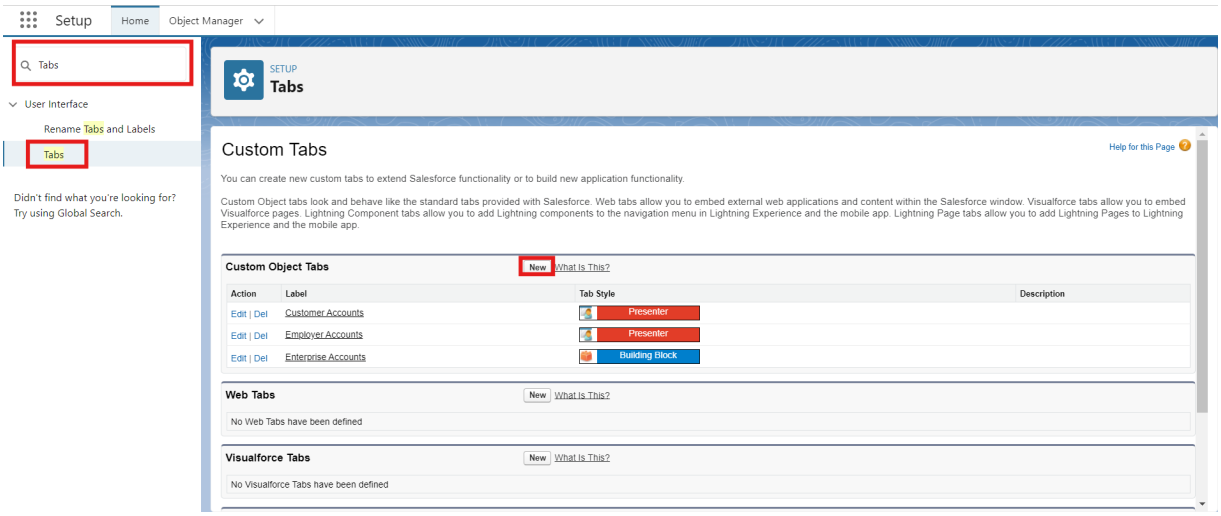
Activity 3 : Enterprise Account Object

1. From the setup page → Click on Object Manager → Click on Create → Click on Custom Object.
    - **Label** : Enterprise Account
    - **Plural Label** : Enterprise Accounts
    - Enter Record Name Label and Format
        Record Name : Enterprise Account Name
        Data Type **:** Text
    - **In Optional features :** check the boxes for Allow Reports | Allow Activities | Track Field History.
    - **Search Status :** check the box for Allow Search.
    - Save

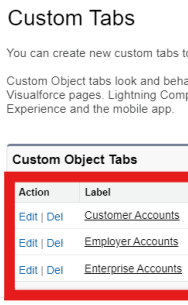
Milestone 3 - Tabs

Activity 1 : Custom tabs creation

1. Go to setup page → type Tabs in Quick Find bar → click on tabs → New (under custom object tab)



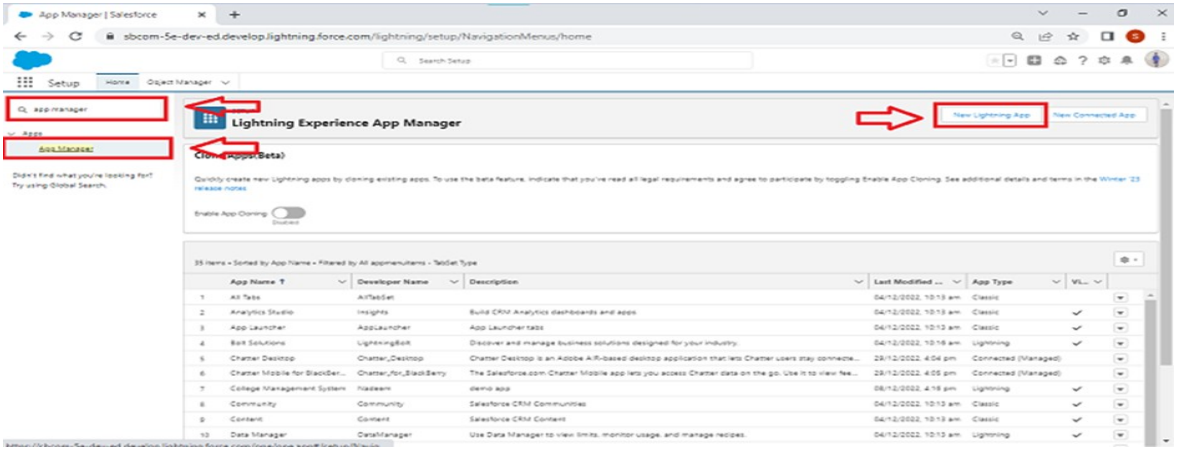
2. Select Object(Employer Accounts) → Select the tab style → Next (Add to profiles page) keep it as default → Next (Add to Custom App) uncheck the include tab .
3. Make sure that the Append tab to users' existing personal customizations is checked.
4. Click save.
5. Repeat the same steps for Customer Accounts and Enterprise Accounts.



Milestone 4 - The Lightning App

Activity 1 : Create a lightning app

1. Go to setup page → search “app manager” in quick find → select “app manager” → click on New lightning App.

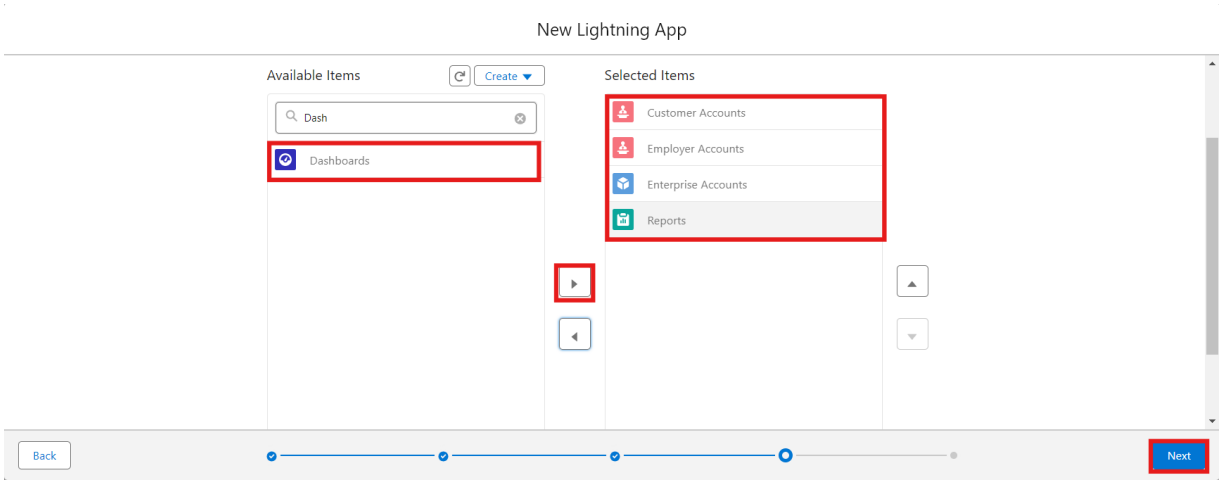


2. Fill the app name in app details and branding as follow
   App Name : TheChaiShop&Co
   Developer Name : Auto Populates
   Add image optional (if you want to give any image you can otherwise not mandatory) - Add Primary color Hex or leave it to default.
   Then click Next.

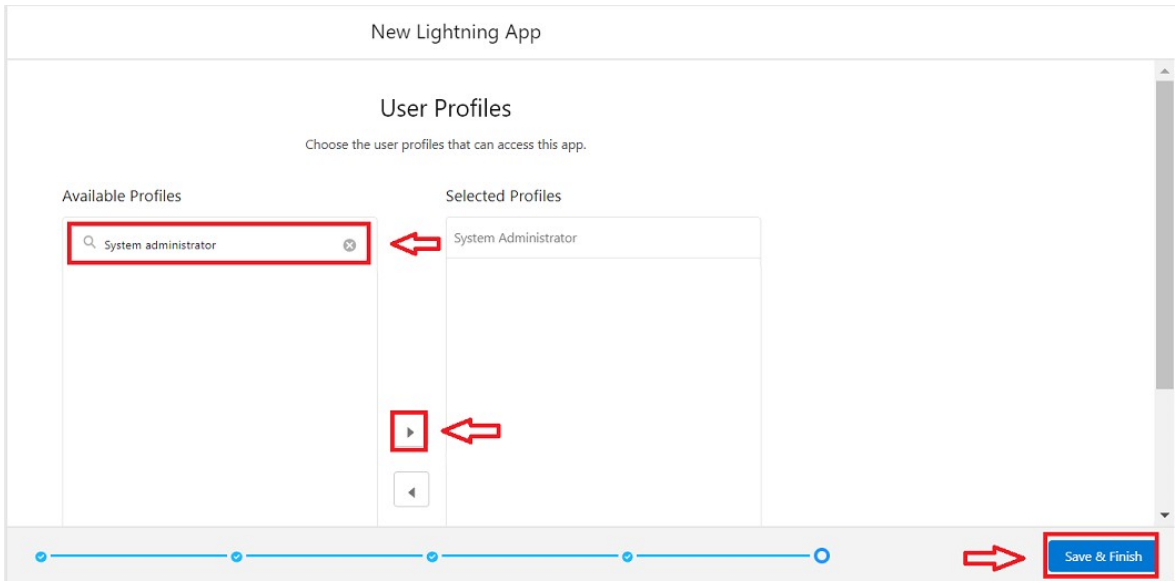
3. In App options
   Navigation style : Standard navigation
   Setup experience : setup
   Supporters form factors : Desktop and phone
   Then click Next

4. In Utility items
   Utility Bar alignment : Default
   Then click Next.

5. Navigation items
   Select the created Custom Objects and required standard objects
   - Customer Accounts
     Employer Accounts
     Enterprise Accounts
     Reports
     Dashboards



6. To Add User Profiles : System Administrator

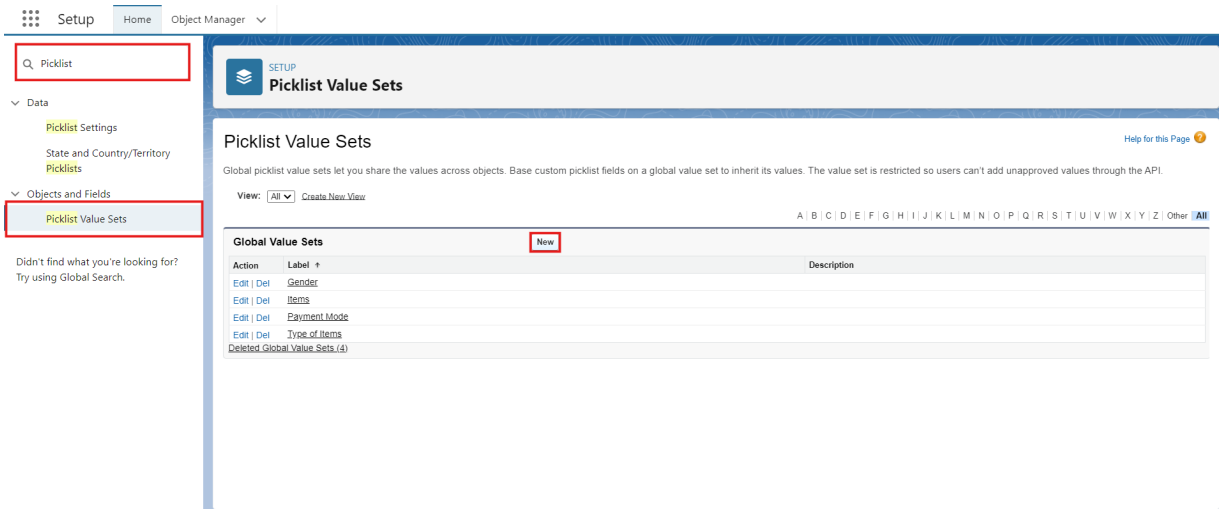


7. Click Save & Finish.

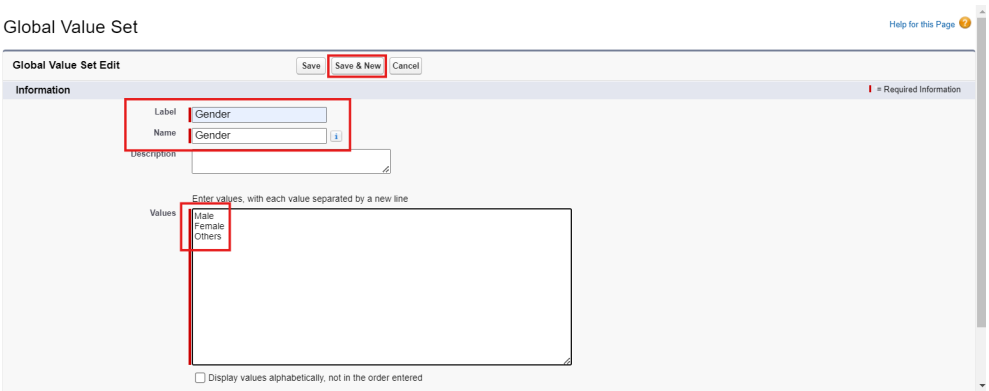
Milestone 5 : Global Picklist

Activity 1 : Create Global PIcklist

1. In Quick find box search for : Picklist value sets and click New.



- **Label** : Gender
  Name : Auto Populates
  Values : Male
  Female
  Others
  Click **Save & New.**



2. Create three more Global picklists following the same steps.
   - **Label** : Items
     Name : Auto populates
     Values : Platinum Tea Powder
     Double Delight Tea Powder
     Super Dust Tea Powder
     Osmania Biscuits (Value Pack)
     Osmania Biscuits (Premium Pack)
     Dry Fruit Biscuits
     Fruit Biscuits
     Salt Biscuits

Multigrain Biscuits
Ragi Biscuits
Double Choco Chip Cookies
Butter Almond Cookies
Melting Moments Cookies
Honey Oats Cookies
Delight Box
Signature Box
Assorted Box

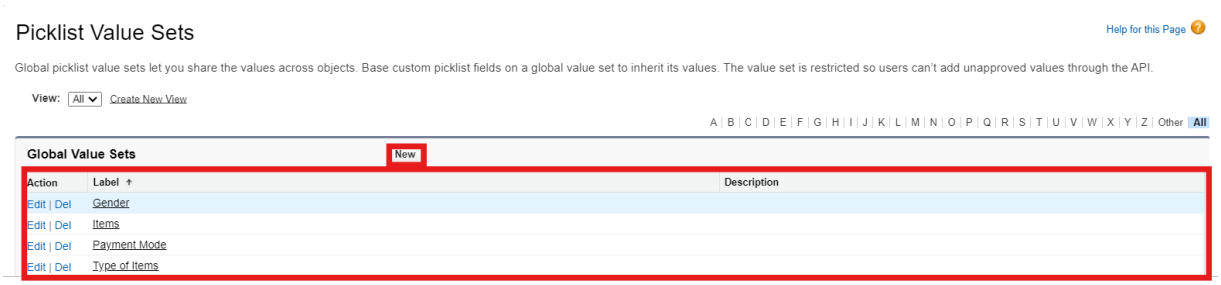
Click **Save & New.**

- **Label** : Payment Mode
  Name : Auto populates
  Values : Gpay
  PhonePe
  Paytm
  Credit or Debit card
  Cash.

  Click **Save & New.**

- **Label :** Type of Items
  Name : Auto populates
  Values : Tea powder
  Cookies
  Gift Hampers

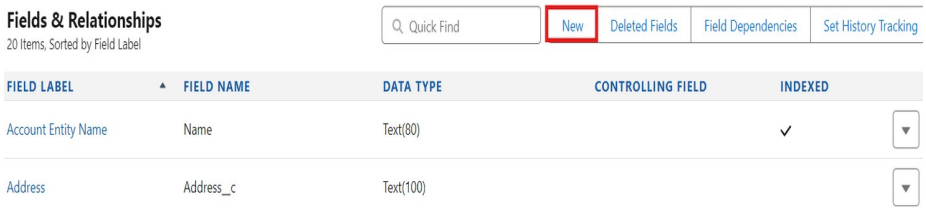
  Click **Save.**



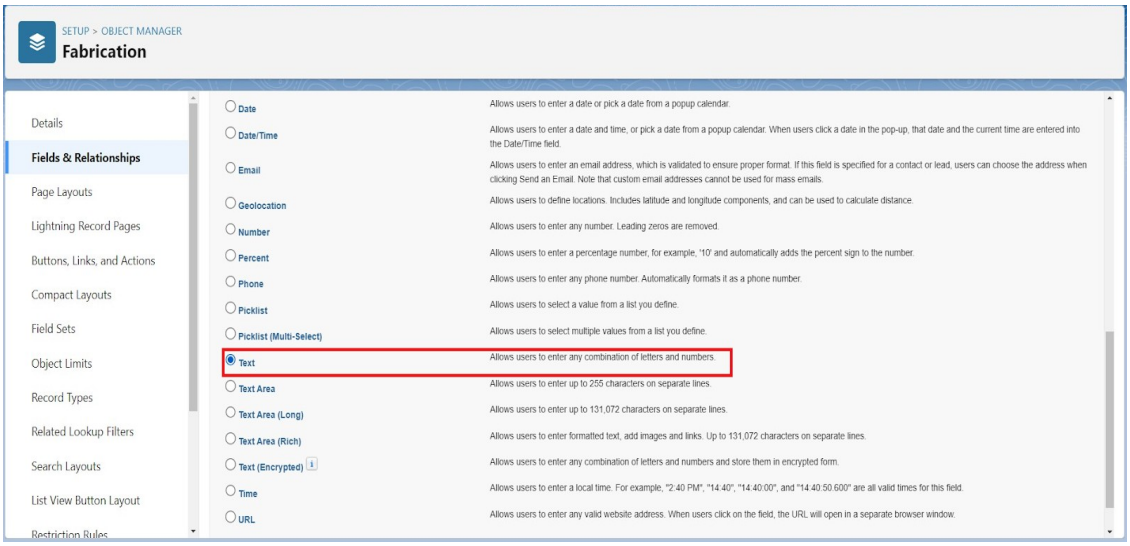
Milestone 6 : Fields & Relationships

Activity 1 : Create Custom fields for Employer Account

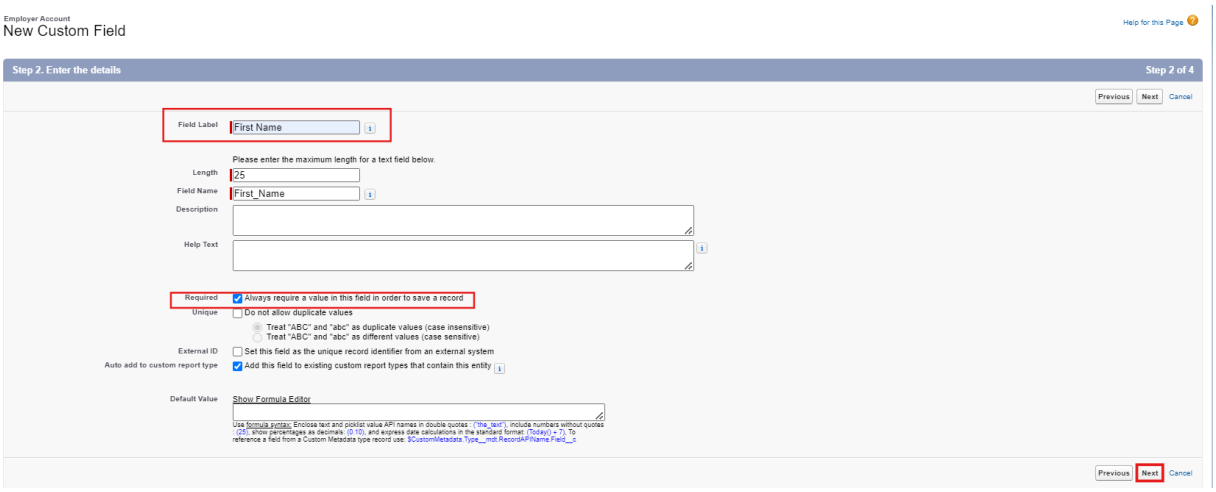
1. In your Salesforce org, click gear icon on the top left and select Setup to open Setup.
2. Click the Object Manager tab. If you don’t see it, enter Object Manager in the Quick Find box.
3. From the object manager page, In the Quick Find box, Search for the custom object you just created : **Employer Account**
4. From the sidebar, click Fields & Relationships. Notice that there are already some fields there. Those are the standard fields.
5. Click New to create a custom field. Tip: Before creating a new field, do a quick search to make sure a similar one doesn’t already exist.



6. Next, choose a data type as Text. Choosing a data type helps you format the field input.



7. Click on next.

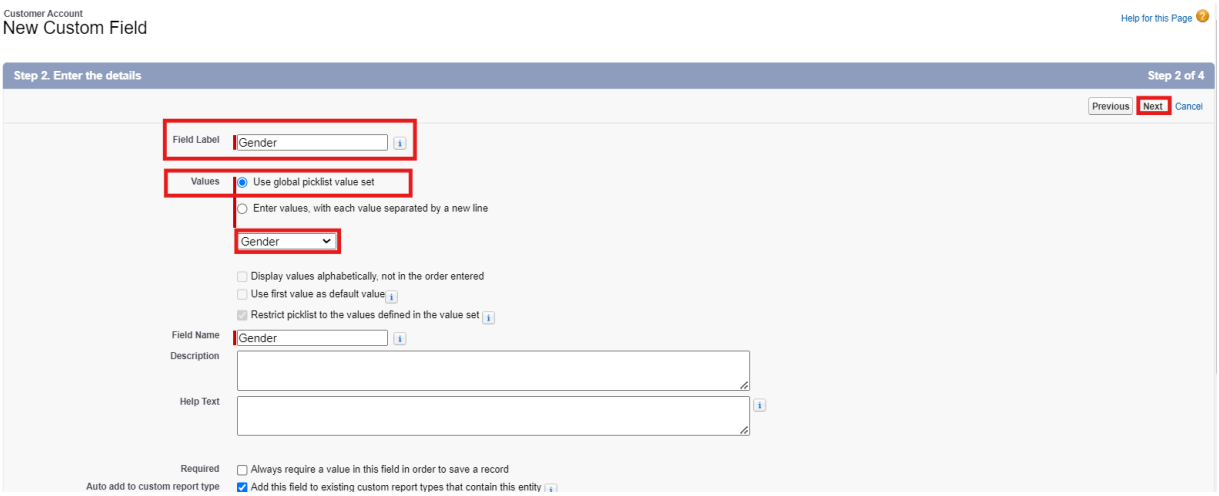


8. Data Type : Text
   Field Label: First Name
   Field Name : Auto Populates
   Length : 25
   Required : Check the box
   Click on Next → Next → Save and new.

9. Repeat the Same steps for remaining fields

   - Data Type : Text
     Field Label: Last Name
     Field Name : Auto Populates
     Length : 25
     Required : Check the box
     Click on Next → Next → Save and new.

   - Data Type : **PickList**
     Field Label: Gender
     Values : Use **Global Picklist** value set and select **Gender** from the dropdown menu.
     Field Name : Auto Populates.
     Required : Check the box
     Click on Next → Next → Save and new.



- Data Type : Phone
  Field Label : Phone Number
  Field Name : Auto Populates
  Required : Check the box
  Click on Next → Next → Save and new.

- Data Type : Email
  Field Label : Email ID
  Field Name : Auto Populates
  Required : Check the box
  Click on Next → Next → Save and new.

- Data Type : Date
  Field Label : Date of Birth
  Field Name : Auto Populates
  Required : check the box
  Click on Next → Next → Save and new.

- Data Type : Text
  Field Label : Aadhar Number
  Field Name : Auto Populates
  Length : 12
  Click on Next → Next → Save and new.

- Data Type : Text
  Field Label : PAN Details
  Field Name : PAN_Details

Length : 10
Click on Next → Next → Save and new.

- Data Type : **Picklist**
  Field Label : Designation
  Field Name : Auto Populates
  Value : Enter value, with each value separated by a new line.
  Manager
  Finance and Accounting
  Kitchen Staff
  Cashier
  Servers
  Click on Next → Next → Save and new.

- Data Type : **Currency**
  Field Label : Salary
  Field Name : Auto Populates
  Click on Next → Next → Save.

Activity 2: Create Custom fields for Customer Account

1. Data Type : Text
   Field Label: First Name
   Field Name : Auto Populates
   Length : 25
   Required : Check the box
   Click on Next → Next → Save and new.

2. Data Type : Text
   Field Label: Last Name
   Field Name : Auto Populates
   Length : 25
   Required : Check the box
   Click on Next → Next → Save and new.

3. Data Type : **PickList**
   Field Label: Gender
   Values : Use **Global Picklist** value set and
   select Gender from the dropdown menu.

Field Name : Auto Populates.
Required : Check the box
Click on Next → Next → Save and new.

4. Data Type : Phone
   Field Label : Phone Number
   Field Name : Auto Populates
   Required : Check the box
   Click on Next → Next → Save and new.

5. Data Type : Date
   Field Label : Date of Birth
   Field Name : Auto Populates
   Required : check the box
   Click on Next → Next → Save and new.

6. Data Type : **Formula**
   Field Label : Date of Birthday is today ?
   Field Name : Auto Populates
   Formula Return : **Checkbox**
   Formula :

```
IF(
AND(
DAY( TODAY() ) = DAY( Date_of_Birth__c ) ,
MONTH ( TODAY()) = MONTH( Date_of_Birth__c )),
TRUE,
FALSE)
```

   Click on Next → Next → Save and new.

7. Data Type : Email
   Field Label : Email ID
   Field Name : Auto Populates
   Required : Check the box
   Click on Next → Next → Save and new.

8. Date Type : Text
   Field Label : Address
   Field Name : Auto Populates
   Length : 125
   Click on Next → Next → Save and new.

9. Date Type : **Multi Select Picklist**
   Field Label : Type of Items
   Values : Use **Global Picklist** value set and
   select **Type of Items** from the dropdown menu.
   Field Name : Auto Populates.
   Required : Check the box
   Click on Next → Next → Save and new.

10. Date Type : **Multi Select Picklist**
    Field Label : Products purchased
    Values : Use **Global Picklist** value set and
    select **Items** from the dropdown menu.
    Field Name : Auto Populates.
    Required : Check the box
    Click on Next → Next → Save and new.

11. Date Type : **Multi Select Picklist**
    Field Label : Chai
    Value : Enter value, with each value separated by a new line.
    TheChaiShop&Co Special Tea
    Single Tea
    Masala Tea
    Lemon Tea
    Ginger Tea
    Green Tea
    Sugar Free Tea
    Click on Next → Next → Save and new.

12. Data Type : **Picklist**
    Field Label : Payment Mode
    Values : Use **Global Picklist** value set and
    select **Payment Mode** from the dropdown menu.
    Field Name : Auto Populates.
    Required : Check the box
    Click on Next → Next → Save and new.

13. Data Type : **Lookup Relationship**
    Related To : Employer Account
    Field Label : Associated Employer

Field Name : Auto Populates

Click on Next → Next → Save and new.

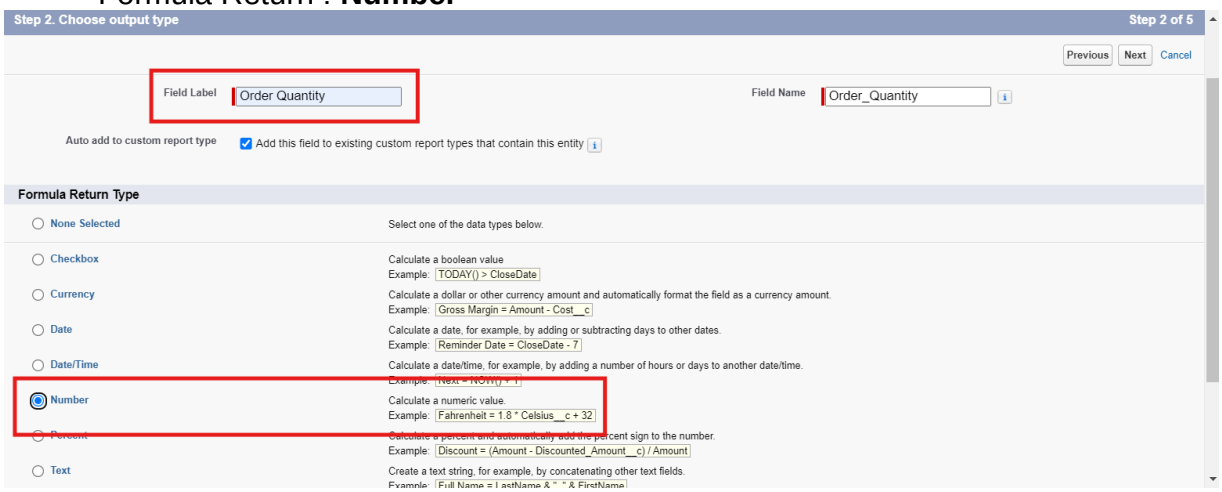


14. Data Type : **Formula**

Field Label : Order Quantity

Field Name : Auto Populates

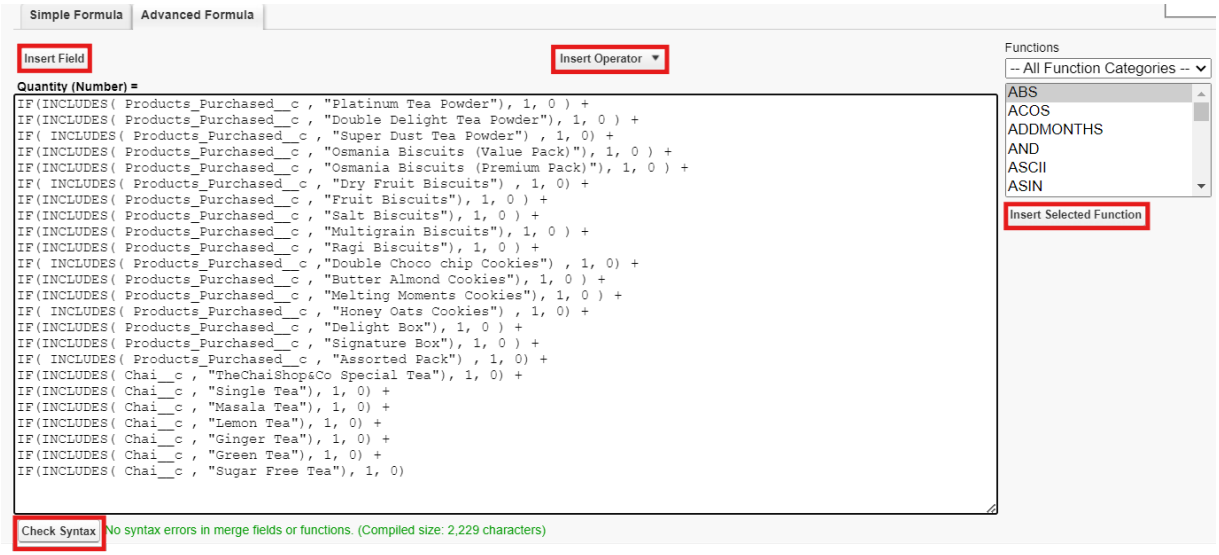
Formula Return : **Number**



Enter Formula = Quantity (Number) =

IF(INCLUDES(Products_Purchased__c , "Platinum Tea Powder"), 1, 0) +
IF(INCLUDES(Products_Purchased__c , "Double Delight Tea Powder"), 1, 0) +
IF(INCLUDES(Products_Purchased__c , "Super Dust Tea Powder") , 1, 0) +
IF(INCLUDES(Products_Purchased__c , "Osmania Biscuits (Value Pack)"), 1, 0) +
IF(INCLUDES(Products_Purchased__c , "Osmania Biscuits (Premium Pack)"), 1, 0) +
IF(INCLUDES(Products_Purchased__c , "Dry Fruit Biscuits") , 1, 0) +
IF(INCLUDES(Products_Purchased__c , "Fruit Biscuits"), 1, 0) +
IF(INCLUDES(Products_Purchased__c , "Salt Biscuits"), 1, 0) +
IF(INCLUDES(Products_Purchased__c , "Multigrain Biscuits"), 1, 0) +
IF(INCLUDES(Products_Purchased__c , "Ragi Biscuits"), 1, 0) +
IF(INCLUDES(Products_Purchased__c ,"Double Choco chip Cookies") , 1, 0) +
IF(INCLUDES(Products_Purchased__c , "Butter Almond Cookies"), 1, 0) +
IF(INCLUDES(Products_Purchased__c , "Melting Moments Cookies"), 1, 0) +
IF(INCLUDES(Products_Purchased__c , "Honey Oats Cookies") , 1, 0) +
IF(INCLUDES(Products_Purchased__c , "Delight Box"), 1, 0) +
IF(INCLUDES(Products_Purchased__c , "Signature Box"), 1, 0) +
IF(INCLUDES(Products_Purchased__c , "Assorted Pack") , 1, 0) +
IF(INCLUDES(Chai__c , "TheChaiShop&Co Special Tea"), 1, 0) +

IF(INCLUDES(Chai__c , "Single Tea"), 1, 0) +
IF(INCLUDES(Chai__c , "Masala Tea"), 1, 0) +
IF(INCLUDES(Chai__c , "Lemon Tea"), 1, 0) +
IF(INCLUDES(Chai__c , "Ginger Tea"), 1, 0) +
IF(INCLUDES(Chai__c , "Green Tea"), 1, 0) +
IF(INCLUDES(Chai__c , "Sugar Free Tea"), 1, 0)



Click on Next → Next → Save and new.

15. Data Type : **Formula**
    Field Label : Total Bill
    Field Name : Auto Populates
    Formula Return : **Number**
    Enter Formula = Quantity (Number) =

IF(INCLUDES(Products_Purchased__c , "Platinum Tea Powder"), 175, 0) +
IF(INCLUDES(Products_Purchased__c , "Double Delight Tea Powder"), 150, 0) +
IF(INCLUDES(Products_Purchased__c , "Super Dust Tea Powder") , 125, 0) +
IF(INCLUDES(Products_Purchased__c , "Osmania Biscuits (Value Pack)"), 50, 0) +
IF(INCLUDES(Products_Purchased__c , "Osmania Biscuits (Premium Pack)"), 80, 0) +
IF(INCLUDES(Products_Purchased__c , "Dry Fruit Biscuits") , 110, 0) +
IF(INCLUDES(Products_Purchased__c , "Fruit Biscuits"), 160, 0) +
IF(INCLUDES(Products_Purchased__c , "Salt Biscuits"), 110, 0) +
IF(INCLUDES(Products_Purchased__c , "Multigrain Biscuits"), 170, 0) +
IF(INCLUDES(Products_Purchased__c , "Ragi Biscuits"), 190, 0) +
IF(INCLUDES(Products_Purchased__c ,"Double Choco chip Cookies") , 200, 0) +
IF(INCLUDES(Products_Purchased__c , "Butter Almond Cookies"), 110, 0) +
IF(INCLUDES(Products_Purchased__c , "Melting Moments Cookies"), 150, 0) +
IF(INCLUDES(Products_Purchased__c , "Honey Oats Cookies") , 160, 0) +
IF(INCLUDES(Products_Purchased__c , "Delight Box"), 850, 0) +
IF(INCLUDES(Products_Purchased__c , "Signature Box"), 1500, 0) +
IF(INCLUDES(Products_Purchased__c , "Assorted Pack") , 600, 0) +
IF(INCLUDES(Chai__c , "TheChaiShop&Co Special Tea"), 70, 0) +
IF(INCLUDES(Chai__c , "Single Tea"), 50, 0) +
IF(INCLUDES(Chai__c , "Masala Tea"), 60, 0) +
IF(INCLUDES(Chai__c , "Lemon Tea"), 40, 0) +

IF(INCLUDES(Chai__c , "Ginger Tea"),55, 0) +
IF(INCLUDES(Chai__c , "Green Tea"), 45, 0) +
IF(INCLUDES(Chai__c , "Sugar Free Tea"), 55, 0)



Click on Next → Next → Save.

Activity 3: Create Custom fields for Enterprise Account

1. Data Type : Text
   Field Label: Enterprise Name
   Field Name : Auto Populates
   Length : 25
   Required : Check the box
   Click on Next → Next → Save and new.

2. Data Type : Phone
   Field Label : Phone Number
   Field Name : Auto Populates
   Required : Check the box
   Click on Next → Next → Save and new.

3. Data Type : Email
   Field Label : Email ID
   Field Name : Auto Populates
   Required : Check the box
   Click on Next → Next → Save and new.

4. Date Type : Text

Field Label : Address
Field Name : Auto Populates
Length : 125
Click on Next → Next → Save and new.

5. Date Type : **Multi Select Picklist**
   Field Label : Type of Items
   Values : Use **Global Picklist** value set and
   select **Type of Items** from the dropdown menu.
   Field Name : Auto Populates.
   Required : Check the box
   Click on Next → Next → Save and new.

6. Date Type : **Multi Select Picklist**
   Field Label : Products purchased
   Values : Use **Global Picklist** value set and
   select **Items** from the dropdown menu.
   Field Name : Auto Populates.
   Required : Check the box
   Click on Next → Next → Save and new.

7. Data Type : **Lookup Relationship**
   Related To : Employer Account
   Field Label : Associated Employer
   Field Name : Auto Populates
   Click on Next → Next → Save and new.

8. Data Type : **Formula**
   Field Label : Order Quantity
   Field Name : Auto Populates
   Formula Return : **Number**
   Enter Formula = Quantity (Number) =

```
IF(INCLUDES( Products_Purchased__c , "Platinum Tea Powder"), 1, 0 ) +
        IF(INCLUDES( Products_Purchased__c , "Double Delight Tea Powder"), 1, 0 ) +
        IF( INCLUDES( Products_Purchased__c , "Super Dust Tea Powder") , 1, 0) +
        IF(INCLUDES( Products_Purchased__c , "Osmania Biscuits (Value Pack)"), 1, 0 ) +
        IF(INCLUDES( Products_Purchased__c , "Osmania Biscuits (Premium Pack)"), 1, 0 ) +
        IF( INCLUDES( Products_Purchased__c , "Dry Fruit Biscuits") , 1, 0) +
        IF(INCLUDES( Products_Purchased__c , "Fruit Biscuits"), 1, 0 ) +
        IF(INCLUDES( Products_Purchased__c , "Salt Biscuits"), 1, 0 ) +
        IF(INCLUDES( Products_Purchased__c , "Multigrain Biscuits"), 1, 0 ) +
        IF(INCLUDES( Products_Purchased__c , "Ragi Biscuits"), 1, 0 ) +
        IF( INCLUDES( Products_Purchased__c ,"Double Choco chip Cookies") , 1, 0) +
        IF(INCLUDES( Products_Purchased__c , "Butter Almond Cookies"), 1, 0 ) +
```

```
IF(INCLUDES( Products_Purchased__c , "Melting Moments Cookies"), 1, 0 ) +
IF( INCLUDES( Products_Purchased__c , "Honey Oats Cookies") , 1, 0) +
IF(INCLUDES( Products_Purchased__c , "Delight Box"), 1, 0 ) +
IF(INCLUDES( Products_Purchased__c , "Signature Box"), 1, 0 ) +
IF( INCLUDES( Products_Purchased__c , "Assorted Pack") , 1, 0)
```

Click on Next → Next → Save and new.

9. Data Type : **Formula**
   Field Label : Total Bill
   Field Name : Auto Populates
   Formula Return : **Number**
   Enter Formula = Quantity (Number) =

```
IF(INCLUDES( Products_Purchased__c , "Platinum Tea Powder"), 175, 0 ) +
IF(INCLUDES( Products_Purchased__c , "Double Delight Tea Powder"), 150, 0 ) +
IF( INCLUDES( Products_Purchased__c , "Super Dust Tea Powder") , 125, 0) +
IF(INCLUDES( Products_Purchased__c , "Osmania Biscuits (Value Pack)"), 50, 0 ) +
IF(INCLUDES( Products_Purchased__c , "Osmania Biscuits (Premium Pack)"), 80, 0 ) +
IF( INCLUDES( Products_Purchased__c , "Dry Fruit Biscuits") , 110, 0) +
IF(INCLUDES( Products_Purchased__c , "Fruit Biscuits"), 160, 0 ) +
IF(INCLUDES( Products_Purchased__c , "Salt Biscuits"), 110, 0 ) +
IF(INCLUDES( Products_Purchased__c , "Multigrain Biscuits"), 170, 0 ) +
IF(INCLUDES( Products_Purchased__c , "Ragi Biscuits"), 190, 0 ) +
IF( INCLUDES( Products_Purchased__c ,"Double Choco chip Cookies") , 200, 0) +
IF(INCLUDES( Products_Purchased__c , "Butter Almond Cookies"), 110, 0 ) +
IF(INCLUDES( Products_Purchased__c , "Melting Moments Cookies"), 150, 0 ) +
IF( INCLUDES( Products_Purchased__c , "Honey Oats Cookies") , 160, 0) +
IF(INCLUDES( Products_Purchased__c , "Delight Box"), 850, 0 ) +
IF(INCLUDES( Products_Purchased__c , "Signature Box"), 1500, 0 ) +
IF( INCLUDES( Products_Purchased__c , "Assorted Pack") , 600, 0)
```

10. Data Type : **Picklist**
    Field Label : Payment Mode
    Values : Use **Global Picklist** value set and
    select **Payment Mode** from the dropdown menu.
    Field Name : Auto Populates.
    Required : Check the box
    Click on Next → Next → Save.

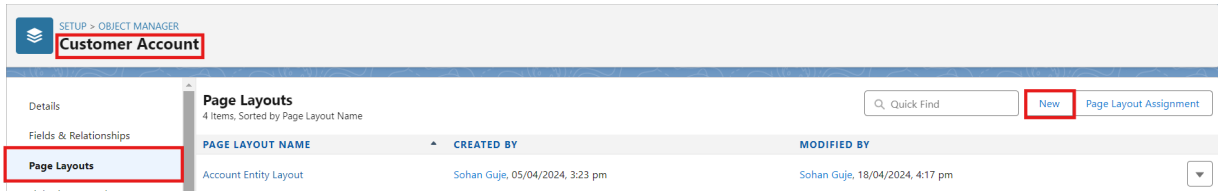
Milestone 7 : Page Layouts

For the personalized field view in Customer Accounts, let's create two page layouts Online orders and Walk-In orders.

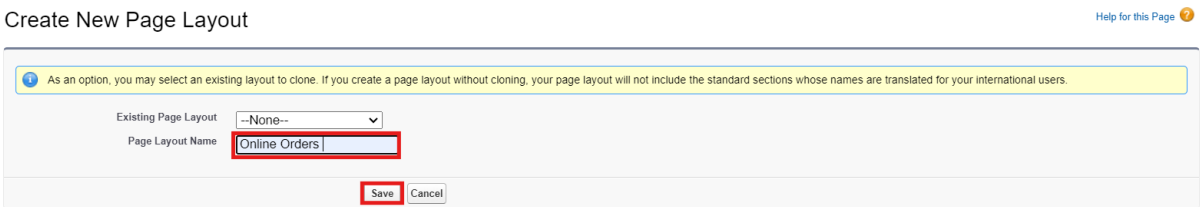
Activity 1: Online Orders page layout

1. Go to the setup page → click on object manager → From drop down click edit for **Customer Account** object.

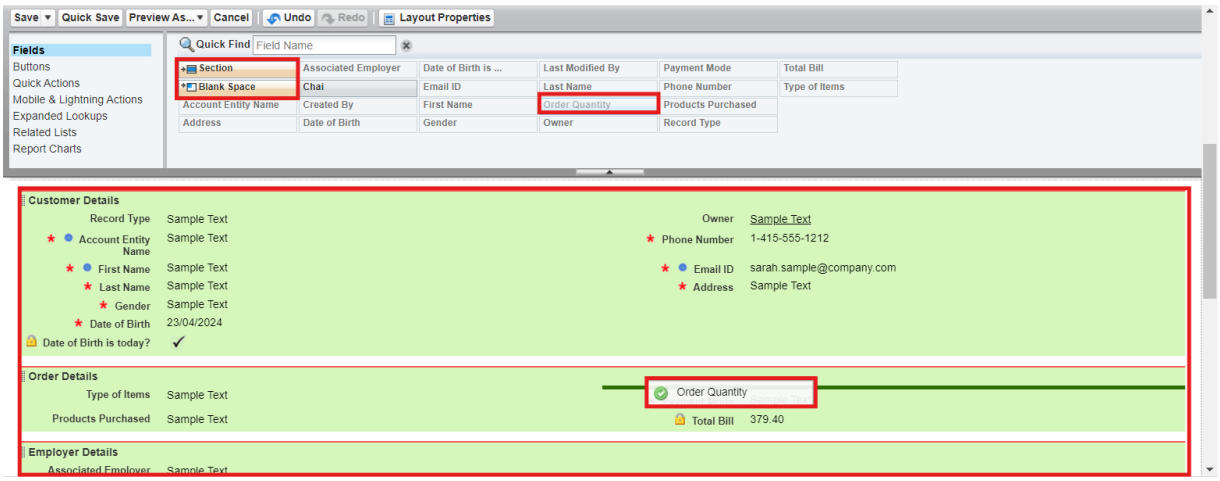
2. Click on the Page Layouts → click New.



3. Page Layout Name : Online Orders and click Save.



4. Drag and Arrange all the required fields as shown below.



5. Click **Save**.

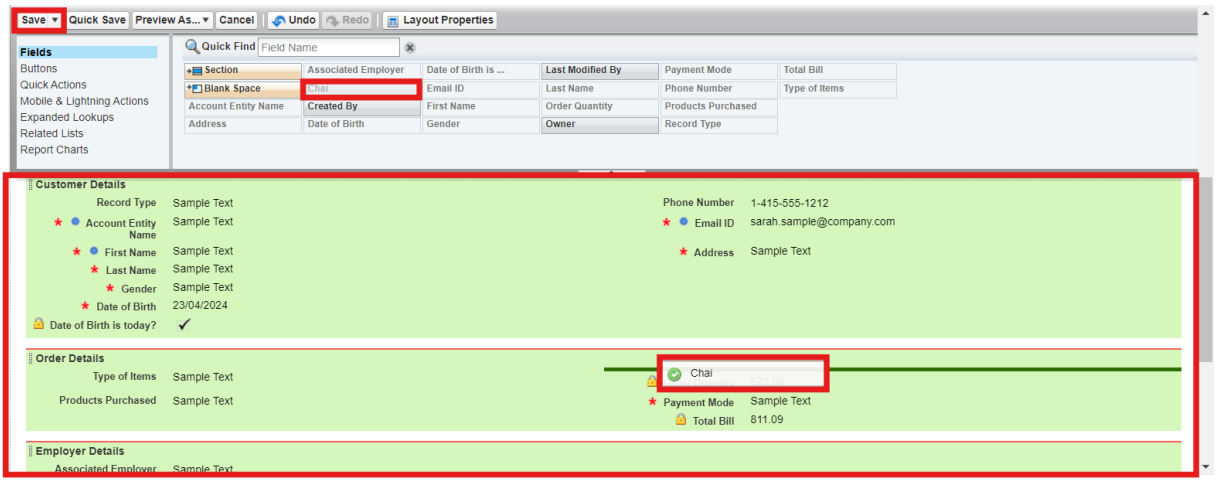
Activity 2: Walk-In Orders page layout

1. Go to the setup page → click on object manager → From drop down click edit for **Customer Account** object.

2. Click on the Page Layouts → click New.

3. Page Layout Name : Walk-In orders and click save.

4. Drag and Arrange all the required fields as shown below.



5. Click **Save**.

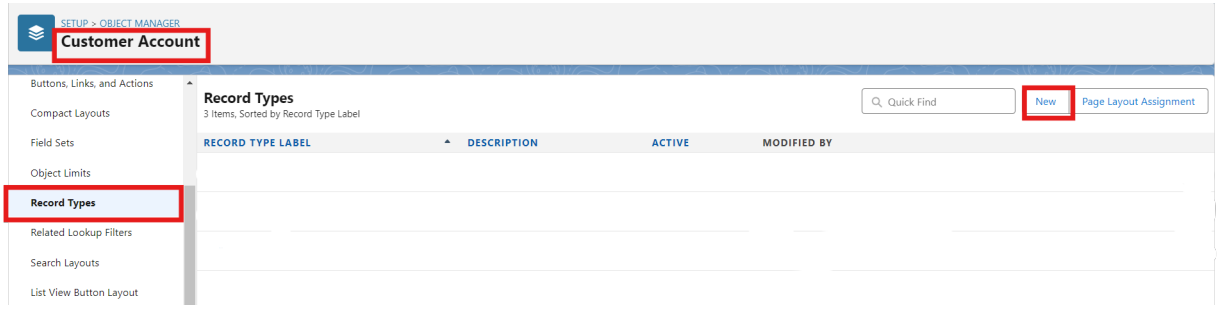
Milestone 8 - Record Types

To Differentiate the records and data stored in Customer Accounts, let's create two different Record Types.
Online orders and Walk-In orders.

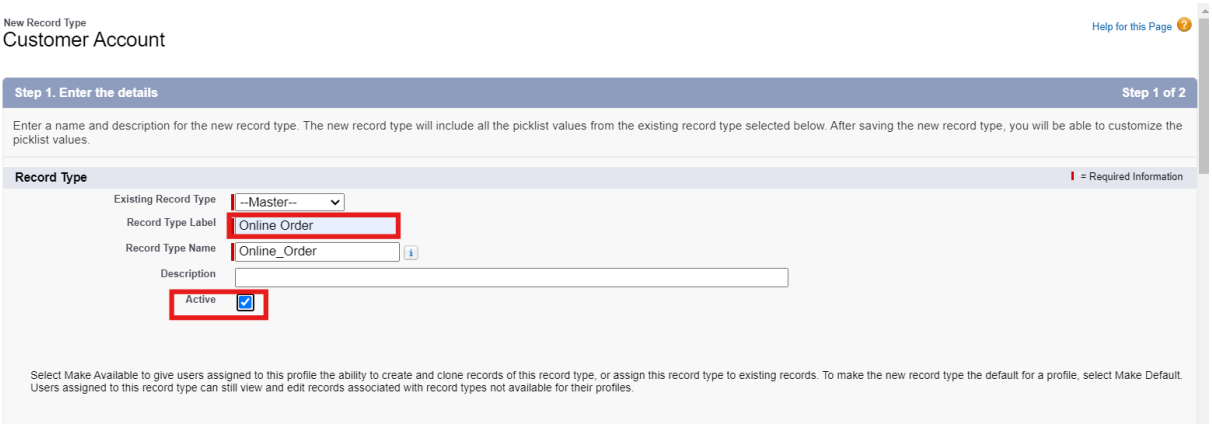
Activity 1: Online Orders Record Type

1. Go to the setup page → click on object manager → From drop down click edit for **Customer Account** object.

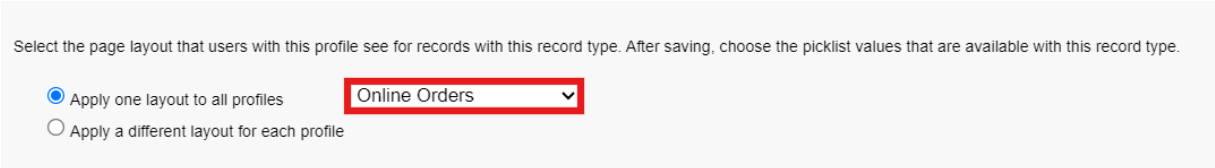
2. Click on the Record Types → click New.



3. Enter Details :
   Existing Record Type : Master
   Record Type Label : Online Order
   Record Type Name : Auto Populates
   Active : Tick the checkbox
   Click Next



4. In Assign Page Layouts
   - Select Apply one layout to all profiles : select **Online orders** from the drop down menu.



5. Click **Save and New**.

Activity 2: Walk-In Orders Record Type

1. Go to the setup page → click on object manager → From drop down click edit for **Customer Account** object.

2. Click on the Record Types → click New.

3. Enter Details :
   Existing Record Type : Master
   Record Type Label : Walk-In Order
   Record Type Name : Auto Populates
   Active : Tick the checkbox
   Click Next

4. In Assign Page Layouts
   - Select Apply one layout to all profiles : select **Walk-In orders** from the drop down menu.

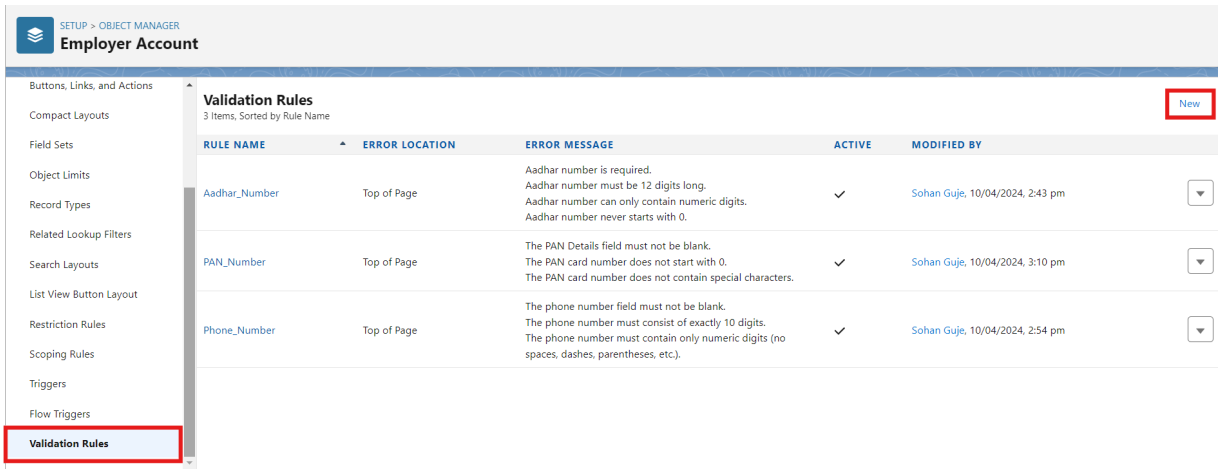
5. Click **Save.**

Milestone 9 - Validation Rules

Activity 1: On Employer Account

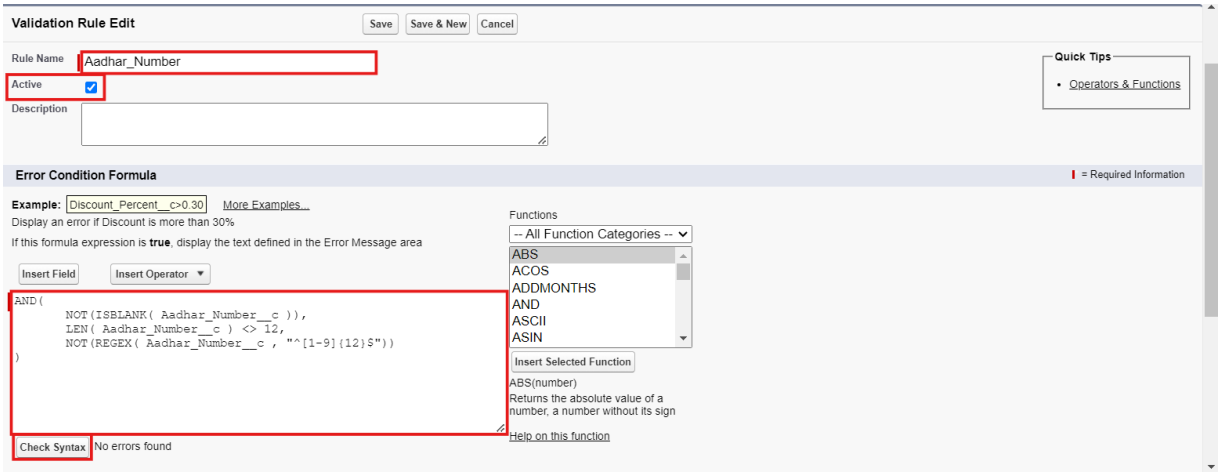
1. Go to the setup page → click on object manager → From drop down click edit for **Employer Account** object.

2. Click on the validation rule → click New.



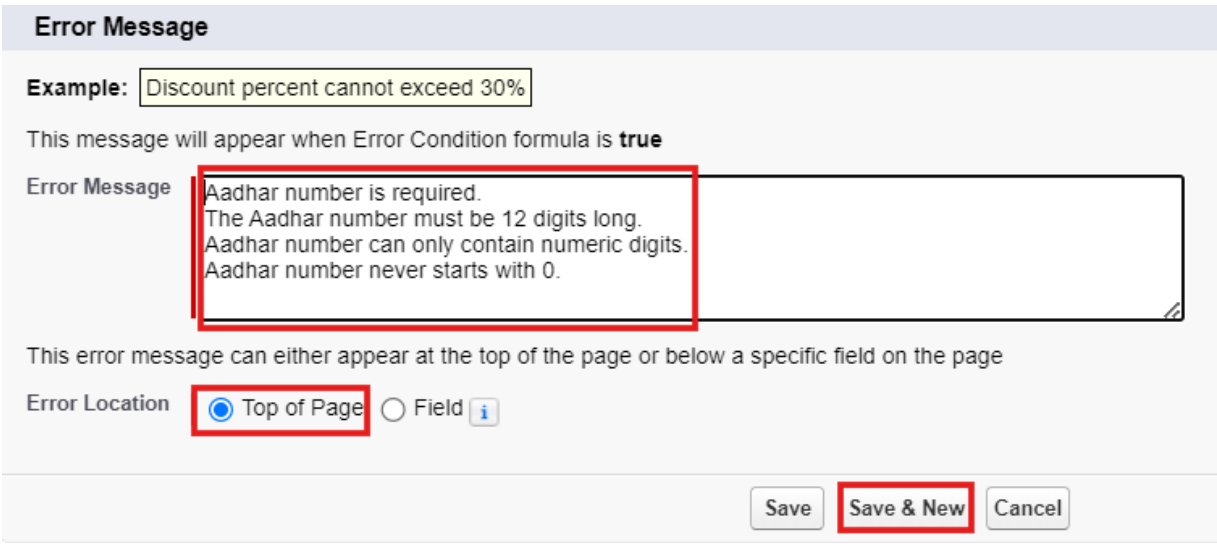
3. **Enter Rule Name :** Aadhar Number
   Error Condition Formula :
   AND(

```
NOT(ISBLANK( Aadhar_Number__c )),
LEN( Aadhar_Number__c ) <> 12,
NOT(REGEX( Aadhar_Number__c , "^[1-9]{12}$"))
)
```



4. **Error Message** : Aadhar number is required.
   The Aadhar number must be 12 digits long.
   Aadhar number can only contain numeric digits.
   Aadhar number never starts with 0

5. Click **Save & New.**



6. Repeat the same steps and create two more validation rules on the employer account object.

- **Enter Rule Name** : PAN Number
  Error Condition Formula :

```
AND(
NOT(ISBLANK( Phone_Number__c )),
NOT(REGEX( Phone_Number__c , "^[1-9]{10}$")),
NOT(REGEX( Phone_Number__c , "^\\d+$"))
)
```

  Error Message : The PAN Details field must not be blank.
  The PAN card number does not start with 0.
  The PAN card number does not contain special characters.

  Click **Save & New.**

- **Enter Rule Name** : Phone Number
  Error Condition Formula :

```
AND(
NOT(ISBLANK( Phone_Number__c )),
NOT(REGEX( Phone_Number__c , "^[1-9]{10}$")),
NOT(REGEX( Phone_Number__c , "^\\d+$"))
)
```

  Error Message :
  The phone number field must not be blank.
  The phone number must consist of exactly 10 digits.
  The phone number must contain only numeric digits
  (no spaces, dashes, parentheses, etc.)

  Click **Save.**

Activity 2: On Customer Account

1. Go to the setup page → click on object manager → From drop down click edit for **Customer Account** object.

2. Click on the validation rule → click New.

3. **Enter Rule Name** : Phone Number

   Error Condition Formula :

```
AND(
NOT(ISBLANK( Phone_Number__c )),
```

```
NOT(REGEX( Phone_Number__c , "^[1-9]{10}$")),
NOT(REGEX( Phone_Number__c , "^\\d+$"))
)
```

Error Message :
The phone number field must not be blank.
The phone number must consist of exactly 10 digits.
The phone number must contain only numeric digits
(no spaces, dashes, parentheses, etc.)

4. Click **Save.**

Activity 3: On Enterprise Account

1. Go to the setup page → click on object manager → From drop down click edit for **Customer Account** object.

2. Click on the validation rule → click New.

3. **Enter Rule Name** : Phone Number

Error Condition Formula :

```
AND(
NOT(ISBLANK( Phone_Number__c )),
NOT(REGEX( Phone_Number__c , "^[1-9]{10}$")),
NOT(REGEX( Phone_Number__c , "^\\d+$"))
)
```

Error Message :
The phone number field must not be blank.
The phone number must consist of exactly 10 digits.
The phone number must contain only numeric digits
(no spaces, dashes, parentheses, etc.)

4. Click **Save.**

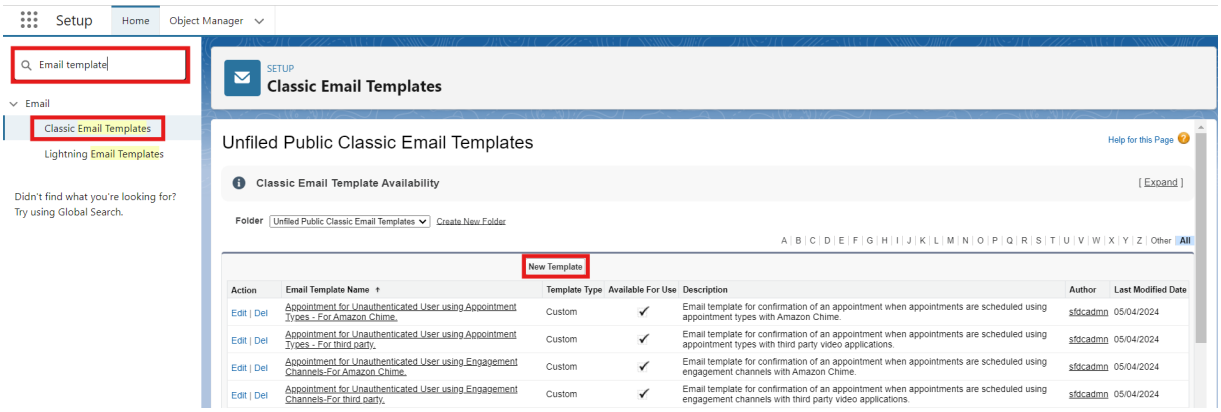
Milestone 10 - Email Template

Activity 1: Email Template

1. Go to setup, In quick find box enter Email template → click on classic Email Template.

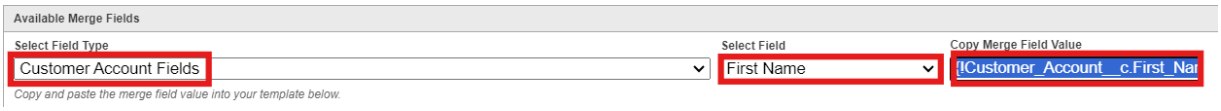
2. Click on→ New Email Template

3. Choose the type of email template you would like to create : **Text** and click **Next**.





- **Available Merge Fields**
  Select Field Type : Customer Accounts Fields
  Select Field : First Name
  Copy Merge Field value : Auto Populates, Copy the value and paste it in the email body where required.



4. Folder : Unfiled public Classic Email templates
5. Click on available for use
6. Email Template Name is “Birthday Offer Email”
7. Template Unique Name : Auto populated
8. Subject : Special Birthday Offer Inside!
9. Email body :

Dear {!Customer_Account__c.First_Name__c},

Happy Birthday from all of us at TheChaiShop&Co! 🎉🎂 We hope your special day is filled with joy, laughter, and wonderful memories.

To celebrate your birthday, we have a special gift just for you! As a token of our appreciation for your continued support, we're delighted to offer you an exclusive birthday discount.

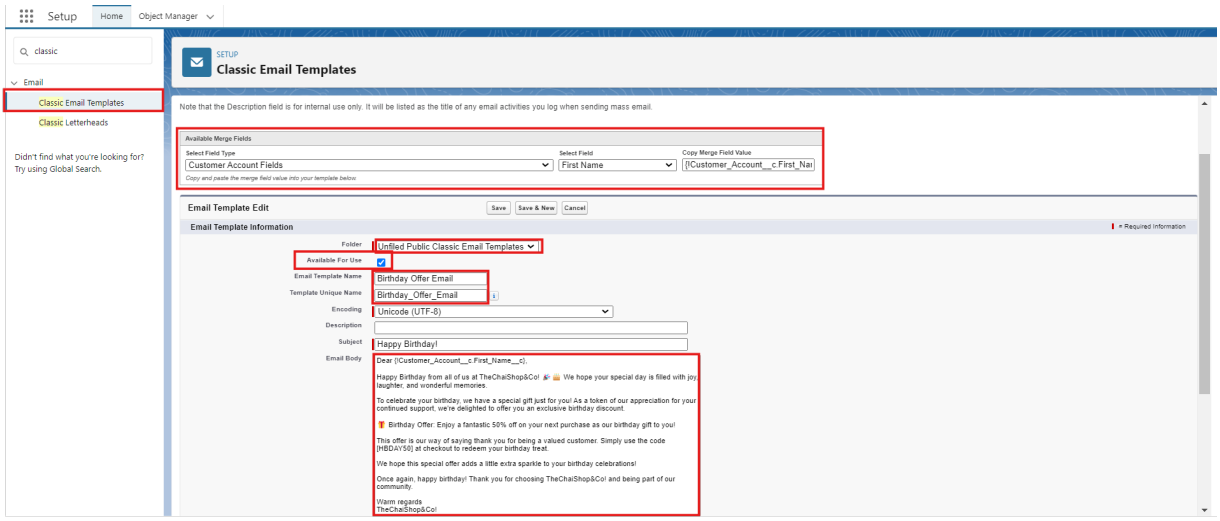
🎁 Birthday Offer: Enjoy a fantastic 50% off on your next purchase as our birthday gift to you!

This offer is our way of saying thank you for being a valued customer. Simply use the code [HBDAY50] at checkout to redeem your birthday treat.

We hope this special offer adds a little extra sparkle to your birthday celebrations!

Once again, happy birthday! Thank you for choosing TheChaiShop&Co! and being part of our community.

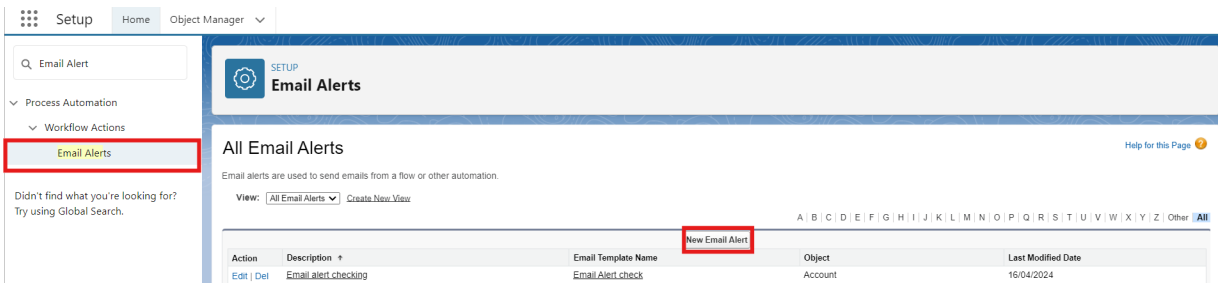
Warm regards
TheChaiShop&Co!



10. Click **Save.**

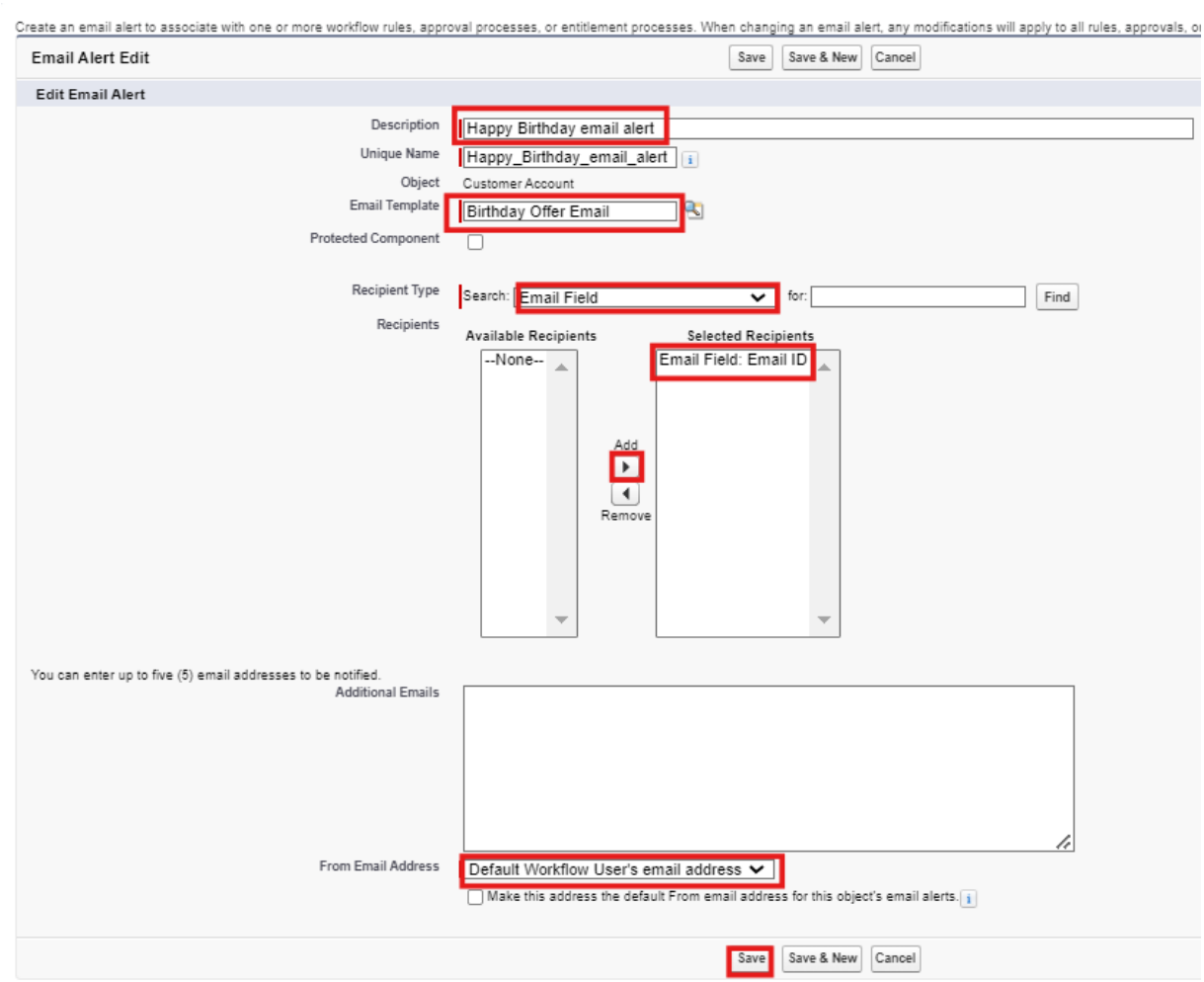
Activity 2: Email Alert

1. Go to setup, In quick find box enter Email Alert → click on Email Alert .



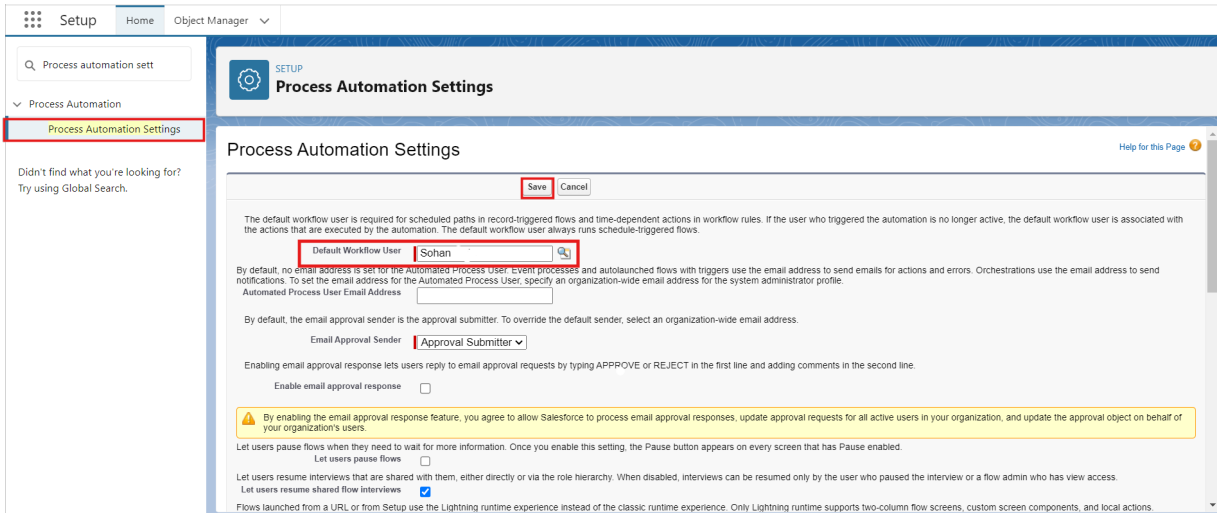
2. Click on New Email Alert.

3. **Description** : Happy Birthday email alert
   Unique Name : Auto Populates
   Object : Customer Account
   Email Template : Search for **Birthday Offer Email**
   Recipient Type : Search : Email field : Email ID
   From Email Address : Default Workflow user's email address
   Click **Save**.



NOTE :

1. Make sure that the Default workflow user is assigned.

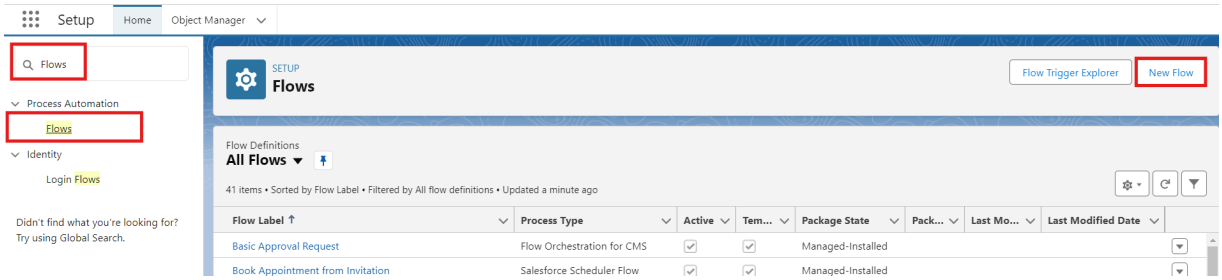


Milestone 11 - Flows

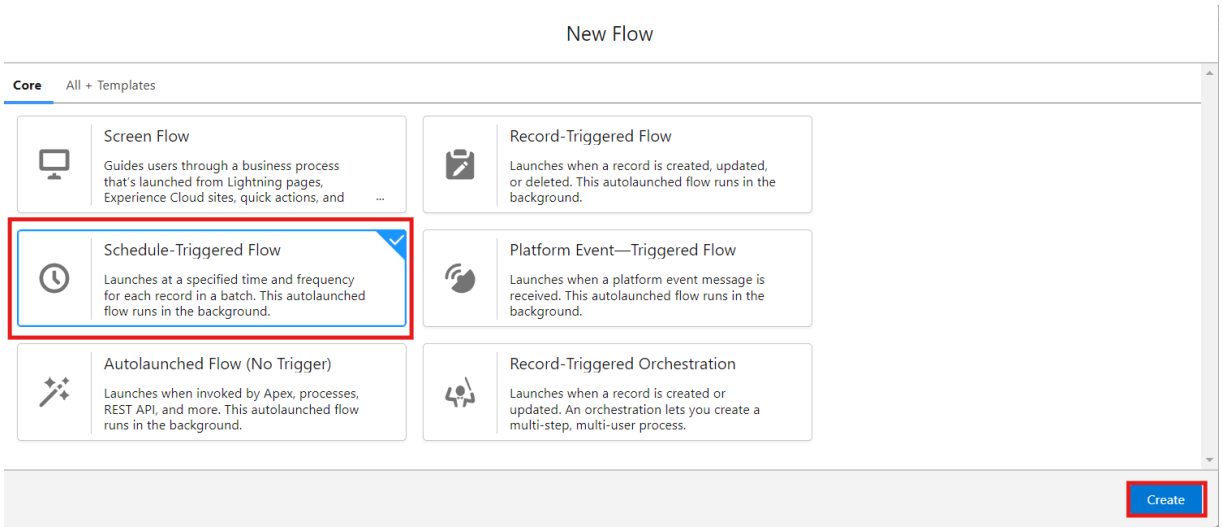
Let's create a scheduled Triggered flow to automate the Birthday offer mails.

Activity 1: Schedule Triggered Flow

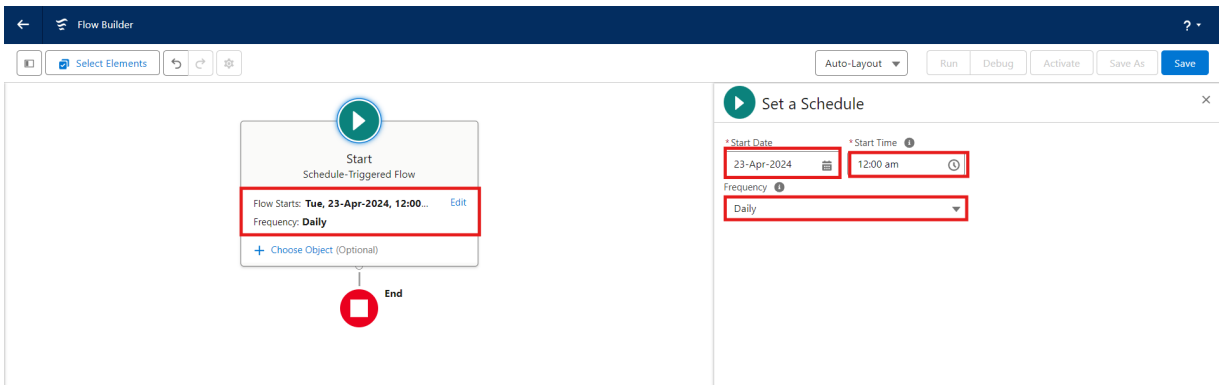
1. Go to setup, Type Flow in quick find box → Click on the Flow and Select the New Flow.



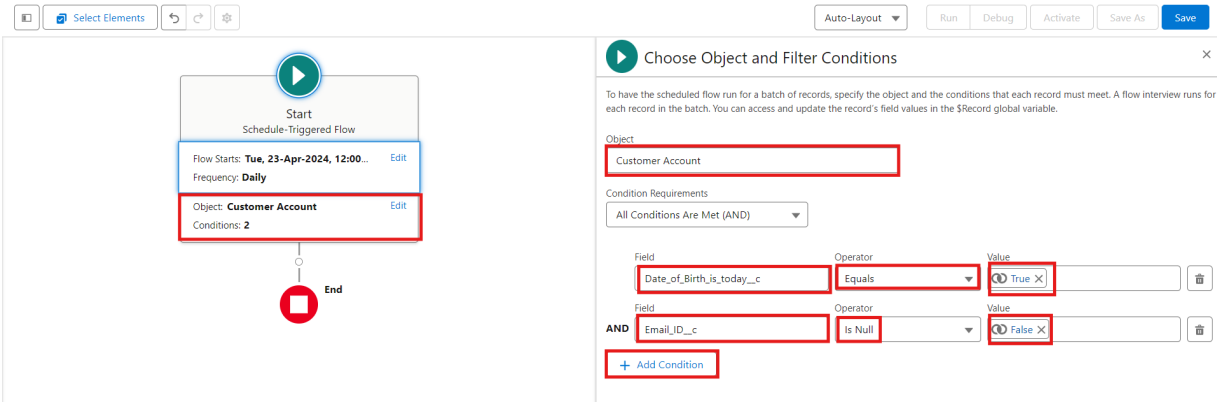
2. Select Schedule - Triggered Flow and click Create.



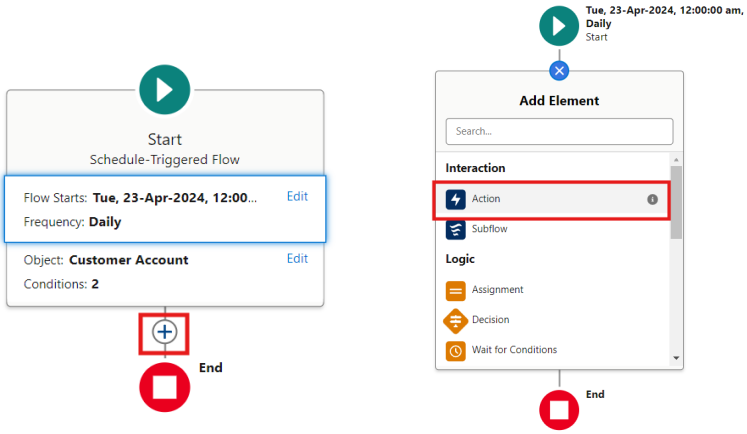
3. **Set a Schedule**
   Start Date : Enter the Date, when you want to start the flow.
   Start Time : Enter the Time, when you want to start the flow.
   Frequency : Daily



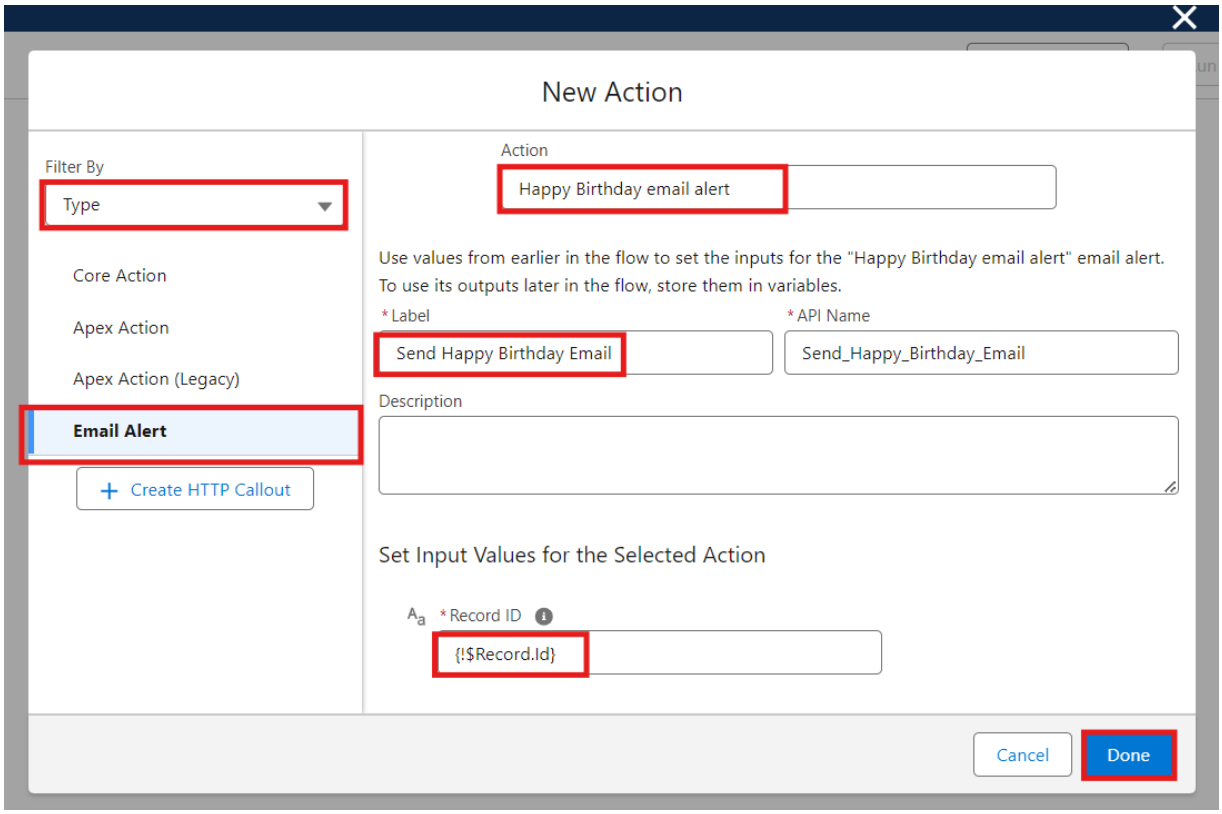
4. **Create Object**
   Object : Customer Account
   Condition Requirements : All conditions Are Met (AND)
   - Field : Date_of_Birthday_is_today__c
     Operator : Equals
     Value : $GlobalConstant.True
   - Field : Email_ID_c
     Operator : IS NULL
     Value : $GlobalConstant.False

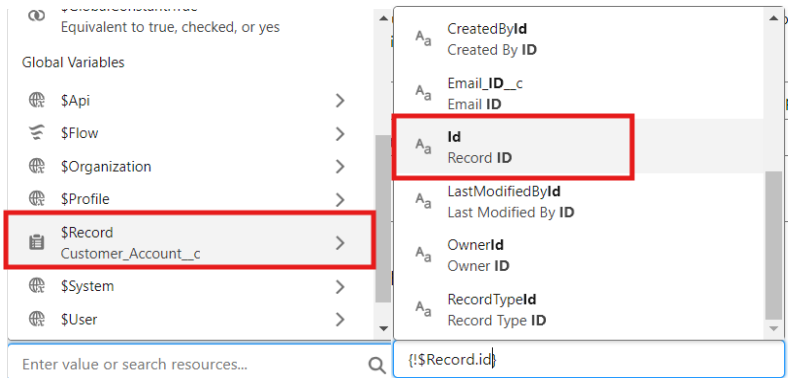


5. Click (+) Icon on the flow builder and Select **Action Component.**



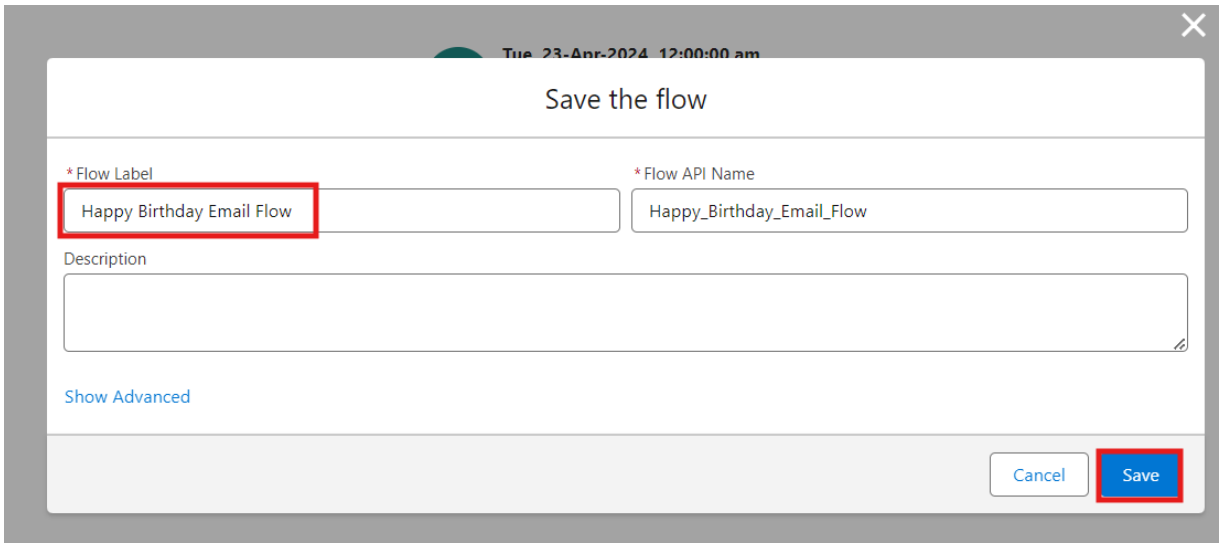
6. In New Action Page
   Filter By : Type and select Email Alert
   In Action : Search and select Happy Birthday email alert
   Label : Send Happy Birthday Email
   API Name : Auto Populates
   Set Input Values for the Selected Action : Record ID : {!$Record.Id}



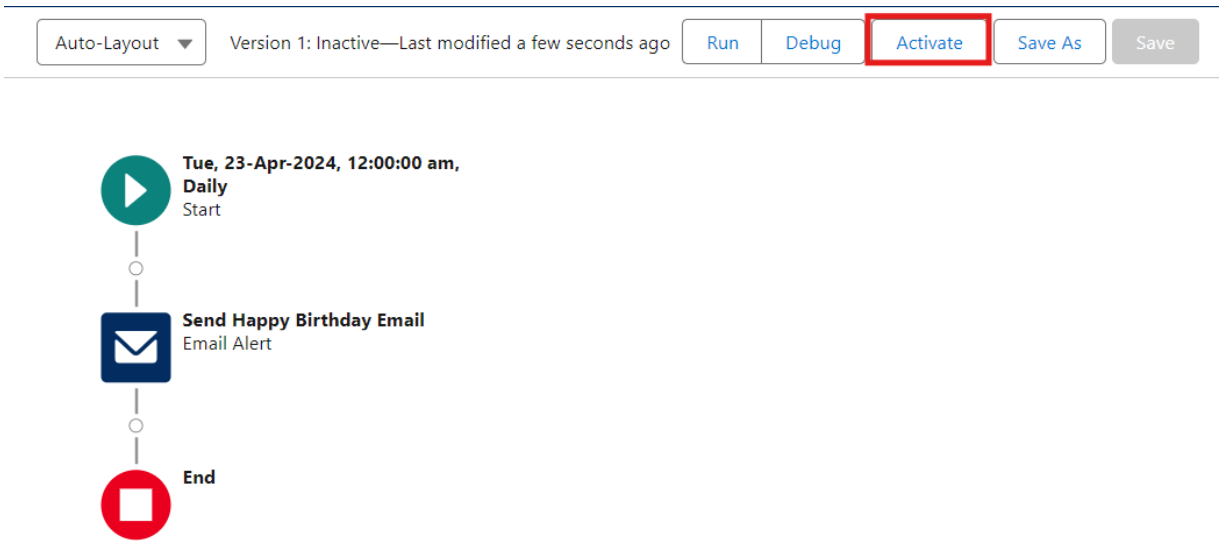


7. Click **Done**.

8. Click Save on the top right
   Flow Label : Happy Birthday Email Flow
   Flow API Name : Auto Populates
   Click **Save.**



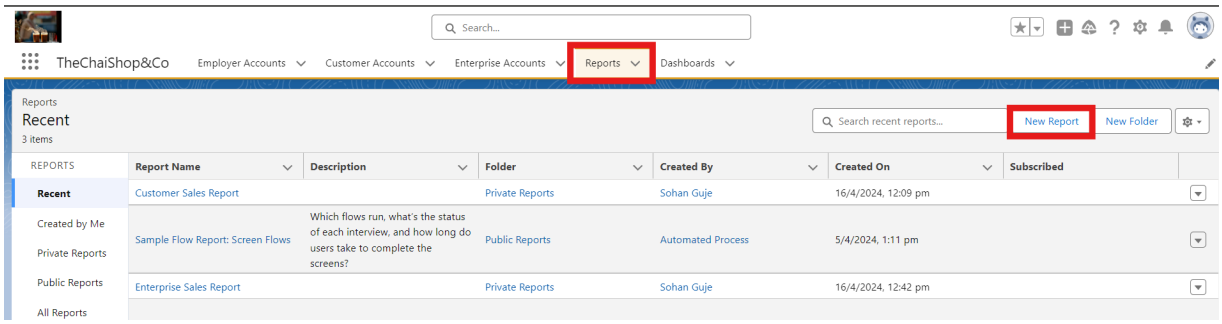
9. Click **Activate**.



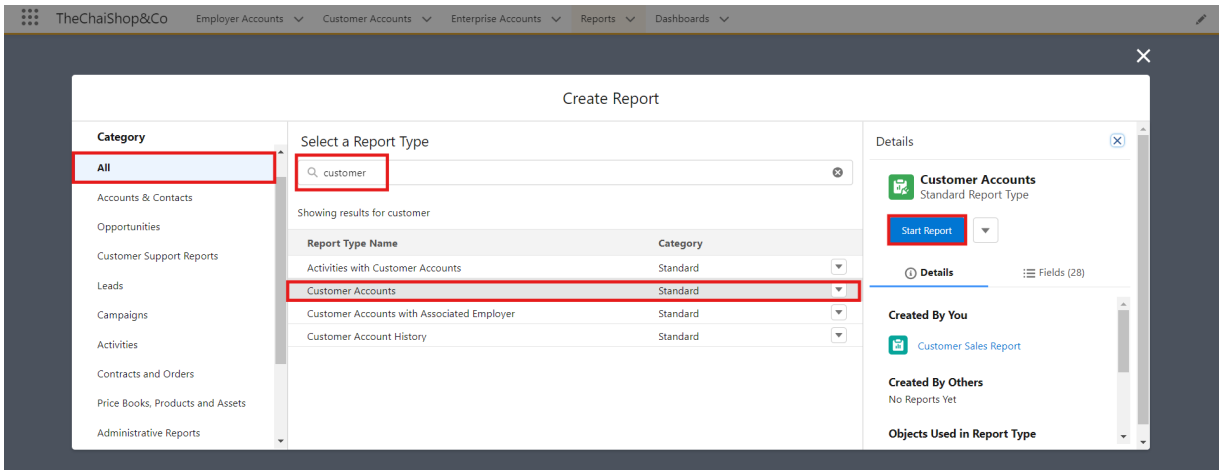
Milestone 12 - Reports & Dashboards

Activity 1: Customer Account Sales Report

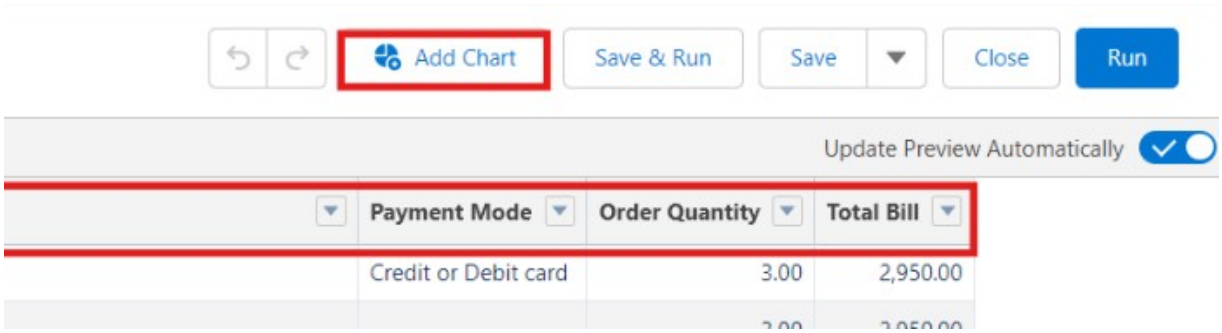
1. Click App Launcher and select TheChaiShop&Co.
2. Click Reports
3. Click New Report

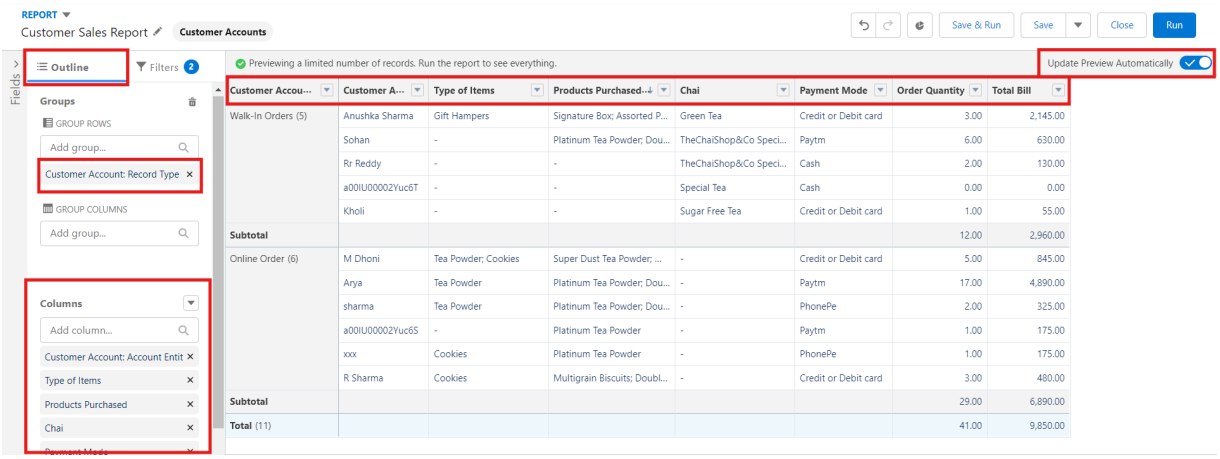


4. In search type Customer Accounts and Select Customer Accounts.
5. Click Start Report.

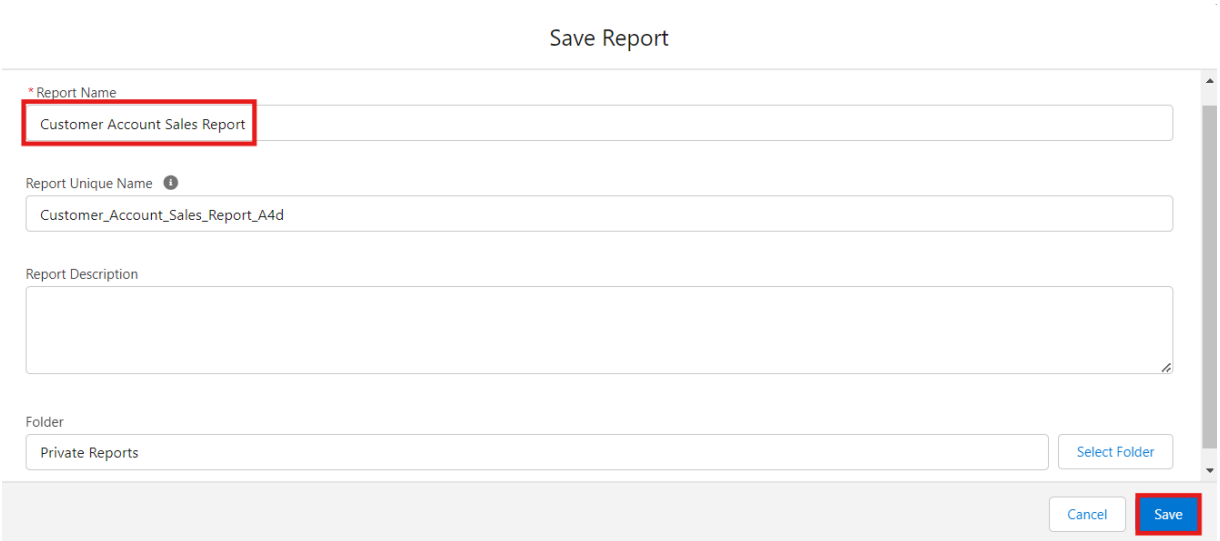


6. Click the toggle **Update Preview Automatically** to **ON**.
7. Drag and Drop all the required Columns from the Outline pane.
8. **Group Rows** : Search and Select
   Customer Account: Record Type
9. Click Add Chart on the top.





10. Click **Save** on the top right.
    Enter
    Report Name: Customer Account Sales Report
    Report Unique Name : Auto Populates
    Click **Save**.

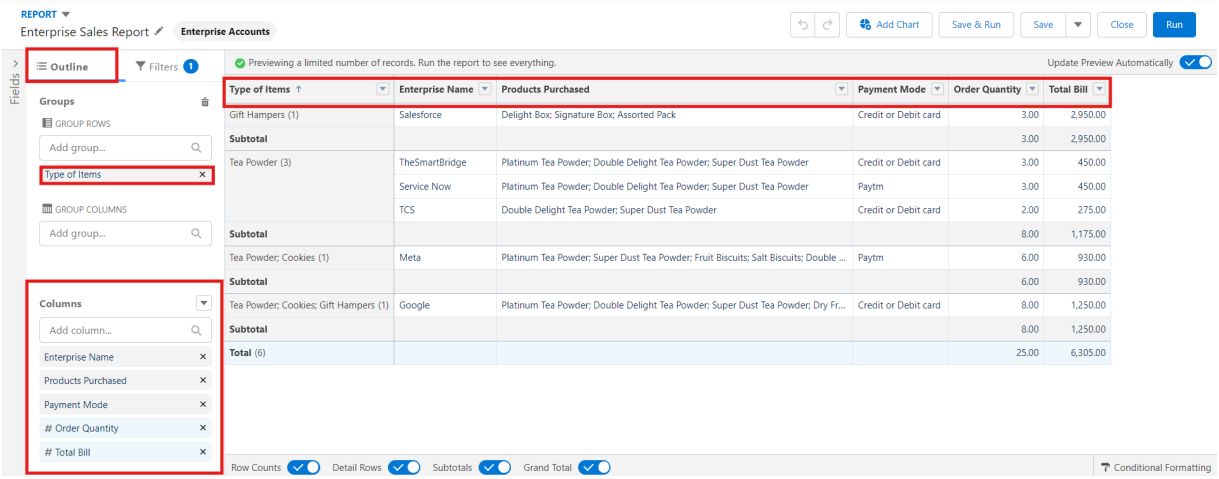


Activity 2: Enterprise Account Sales Report

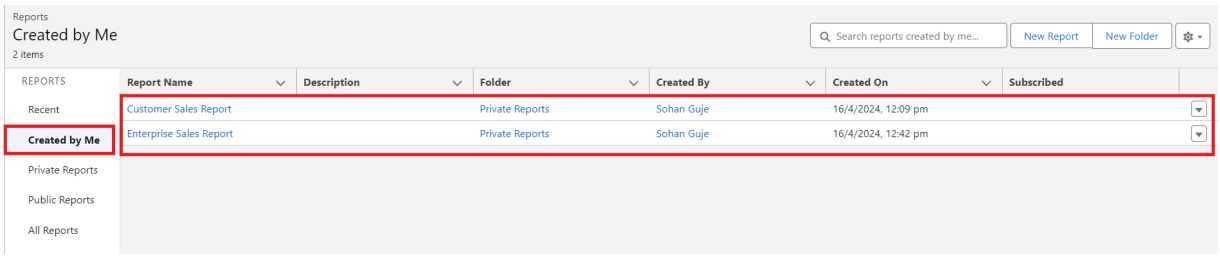
1. Repeat the same steps to create another report.
2. Click Reports
3. Click New Report
4. In search type Enterprise Accounts and Select Enterprise Accounts.
5. Click Start Report.
6. Click the toggle **Update Preview Automatically** to **ON**.
   Drag and Drop all the required Columns from the Outline pane.
   Group Rows : Search and Select

Type of Items.

7. Click Add Chart on the top.

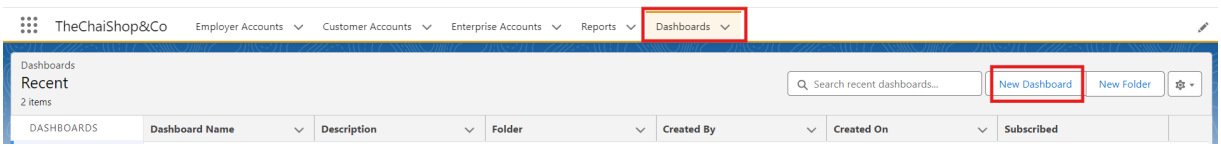


8. Click **Save** on the top right.
    a. Enter
    b. Report Name : Enterprise Account Sales Report
    c. Report Unique Name : Auto Populates
    d. Click **Save**.

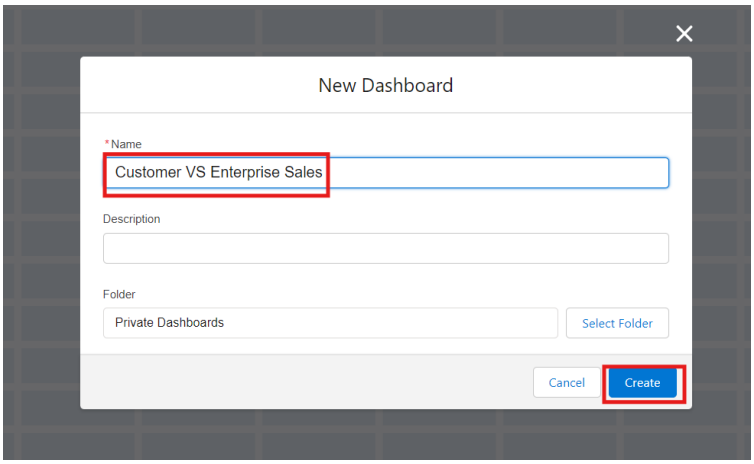


Activity 3: Dashboard

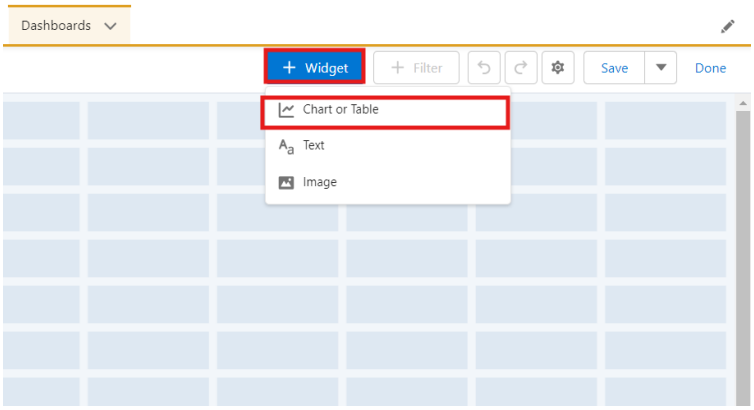
1. Click Dashboards
2. Click New Dashboards



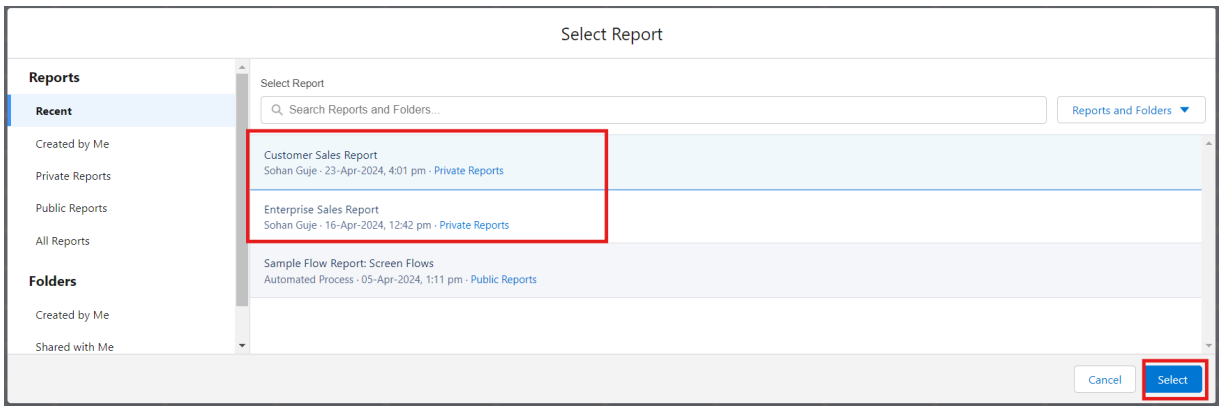
3. Name : Customer Vs Enterprise Sales
4. Click Create.



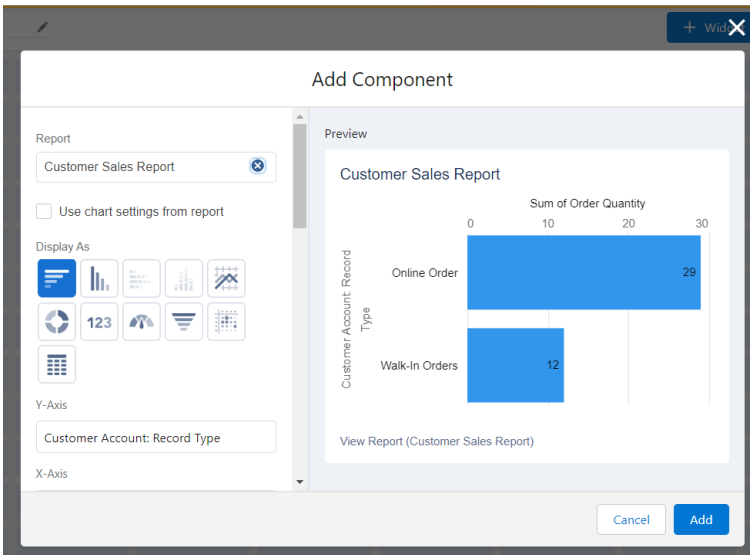
5. Click Widget
6. Select Chart or Table.



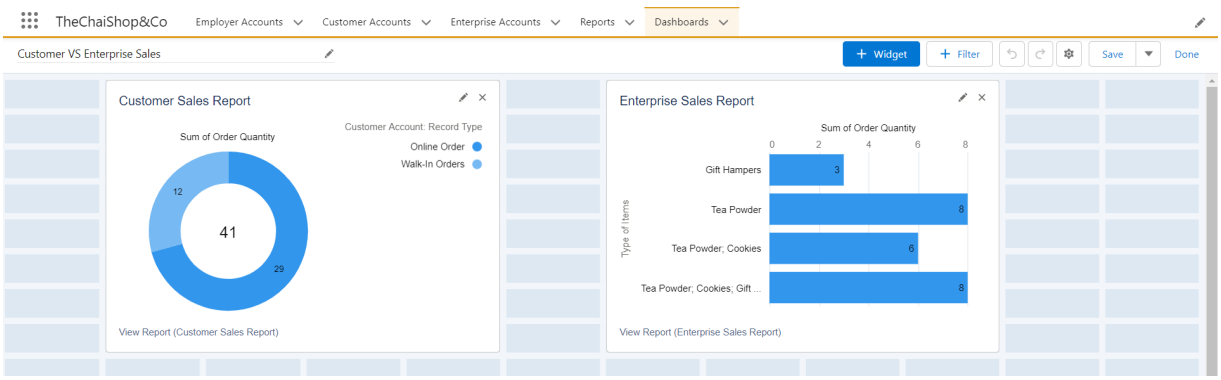
7. Select Customer Account Sales Report



8. Display As : Select the one which you feel comfortable with.
9. Click **Add**.



10. Repeat the same steps for the Enterprise Account Sales Report.
11. Click **Save**.



Milestone 12 - Conclusion

In conclusion, the development of TheChaiShop&Co CRM application marks a significant stride in enhancing the store's operational efficiency and customer relationship management capabilities. Seamlessly integrating employer details, customer information (both online and offline), and enterprise data, this application streamlines business processes and enables informed decision-making. The automation of sales calculations, including order quantity and total bill amount, reduces errors and empowers employees to focus on delivering exceptional service. Moreover, automated email notifications for customers' birthdays foster personalized communication and enhance customer engagement.

7. FUNCTIONAL & PERFORMANCE TESTING

During the testing phase, screenshots were captured from the Salesforce Lightning interface to verify correct system behavior. These include:

- Record creation for Employer, Customer, and Enterprise objects
- Validation rule testing for Aadhar, PAN, and phone numbers
- Verification of record types and page layouts
- Formula field validation for order quantity and billing
- Flow execution for birthday email alerts
- Report and dashboard generation

Performance testing ensured fast response time and system stability during concurrent usage.

8. ADVANTAGES & DISADVANTAGES

Advantages

- Centralized data management
- Reduced manual work

- Automated calculations
- Improved customer engagement
- Real-time analytics

Disadvantages

- Requires Salesforce license
- Internet dependency
- Limited offline access

9. Conclusion:

The CRM Application for TheChaiShop&Co successfully improves business operations by integrating customer data, automating billing, and enabling intelligent reporting. The system reduces errors, saves time, and enhances customer relationships through automation and analytics.

10. FUTURE SCOPE

- Mobile CRM integration
- AI-based sales prediction
- WhatsApp/SMS alerts
- Inventory management
- Multi-branch support
- Advanced customer profiling

11. APPENDIX

Source Code : Salesforce Configuration & Flows

Dataset : Customer and Sales Records

Tools Used : Salesforce Developer Org

Thank You